



# Thermo-Mechanical Simulation Framework for Assessing Process-Induced Delamination Risks in Large-Area Fan-Out Panel-Level Packaging

**Hsien-Chie Cheng**

Department of Aerospace and  
Systems Engineering,  
Feng Chia University,  
Taichung 40724, Taiwan

**Ching-Feng Yu<sup>1</sup>**

Department of Mechanical Engineering,  
National United University,  
Miaoli 360303, Taiwan

*This study develops a finite element modeling (FEM) framework for predicting interfacial fracture behavior during the debonding stage of fan-out panel-level packaging (FOPLP). Although FOPLP offers advantages such as mechanical robustness, high throughput, and cost efficiency, it is susceptible to residual stresses and interfacial delamination, particularly during debonding. To address these reliability concerns, a multiscale FEM approach integrating global and detailed submodels was established to resolve localized stress fields, with an emphasis on regions surrounding critical copper pillars. Simulation results indicate that stress concentrations are most severe at the SiNx–under bump metallurgy (UBM) interface. Parametric studies show that increasing the chamfer radius at UBM corners significantly reduces stress peaks, improving reliability. Furthermore, reducing interfacial fracture toughness within the debonding layer was found to reduce stress in the UBM region, suggesting process-level mitigation strategies. Larger UBM etching angles were found to reduce stress in SiNx and UBM layers. The influence of process-induced residual stresses, including those from chemical shrinkage and thermal mismatch, was systematically evaluated. Critical stress values and temperatures at potential failure sites were identified, providing insight into delamination mechanisms. By incorporating experimental observations and simulation techniques such as the virtual crack closure technique (VCCT), the proposed framework enables accurate prediction of interfacial failure and global warpage in FOPLP assemblies. [DOI: 10.1115/1.4070103]*

**Keywords:** fan-out panel-level packaging, finite element modeling, under bump metallurgy, interfacial delamination, energy release rate

## 1 Introduction

Fan-out wafer-level packaging has become a key technology in advanced semiconductor packaging, offering high input/output density, reduced footprint, and ultrathin form factors essential for smartphones, wearables, and internet of things (IoT) devices [1–7]. By minimizing interconnect lengths, fan-out wafer-level packaging improves electrical and thermal performance, while its design flexibility supports heterogeneous integration of diverse chip types on a single panel. The elimination of costly interposers and adoption of panel-level processing also enhance throughput and reduce costs. Fan-out panel-level packaging (FOPLP) extends these benefits by employing large-area substrates instead of wafers, improving process uniformity and reducing edge-related issues such as thermal mismatch and warpage [8–14]. This scalability makes FOPLP a cost-effective solution for high-volume consumer electronics.

However, challenges remain, particularly during debonding. Local stress concentrations can induce interfacial delamination, compromising thermal and electrical performance. Residual stresses arising from coefficient of thermal expansion (CTE) mismatch and epoxy molding compound (EMC) shrinkage further contribute to warpage and reliability issues.

While warpage has been widely studied [15–30], mechanical response during debonding remains less explored. Cheng and Liu [15] developed warpage simulations incorporating shrinkage, nonlinear effects, and gravity, validated with in-line measurements. Dijk et al. [16] introduced a method for early-stage warpage control and CTE characterization. Yu et al. [17] employed process-aware finite element modeling (FEM) with element birth and death techniques, using response surface optimization to validate warpage suppression strategies. Chen et al. [18] showed that optimizing EMC and substrate thicknesses can reduce warpage in multilayer redistribution layers (RDL)-first processes. Lee et al. [19] used a multilevel equivalent material model to predict fine RDL and bump behavior with less than 5% error. Braun et al. [20] addressed the role of thermal and shrinkage-induced effects through the design of experiments (DOE) and compression molding. Fewer studies focus

<sup>1</sup>Corresponding author.

Contributed by the Electronic and Photonic Packaging Division of ASME for publication in the JOURNAL OF ELECTRONIC PACKAGING. Manuscript received June 2, 2025; final manuscript received October 3, 2025; published online December 15, 2025. Assoc. Editor: Willem Van Driel.

on interfacial behavior during debonding. Conversion et al. [21] used FEM and virtual crack closure techniques (VCCT) to analyze delamination, identifying the CTE of EMC above  $T_g$  as critical, with a safe upper limit of 42.56 ppm/°C. Wu et al. [22] developed a creep model incorporating time, temperature, and humidity to predict polyimide (PI) delamination, achieving over 80% accuracy.

To begin the debonding process, a debonding layer and a PI layer are first applied to the carrier substrate. During thermal processing, the PI layer provides dimensional stability. Subsequently, the debonding layer is removed by mechanical, laser, or air pressure methods to release the package structure, as illustrated in Fig. 1. This method minimizes damage to the underlying substrate and devices, while taking advantage of the thermal resilience of PI. Since crack-tip stresses exhibit singular behavior, traditional FEM stress analysis is insufficient for predicting interface failure. Instead, interface crack propagation is evaluated using the strain energy release rate ( $G$ ), introduced by Irwin [31–33] and refined by Rybicki and Kanninen [34]. Shivakumar et al. [35] extended this to three dimensions using orthogonal grids, later improved by Okada et al. [36] and Smith and Raju [37] through nonorthogonal formulations for computing mode-separated energy release rates ( $G_I, G_{II}, G_{III}$ ). The total energy release rate,  $G_T = G_I + G_{II} + G_{III}$ , is used to define interfacial fracture toughness ( $G_C$ ) without spurious oscillations. Mode mixity, expressed via ratios such as  $K_{II}/K_I$  and  $K_I/K_{III}$ , provides insight into the dominance of different failure modes, as explored by Cho et al. [38].

To address the key limitations identified in the literature—namely, (1) a primary focus on warpage while the mechanical response during the critical debonding stage remains less explored, (2) the insufficiency of traditional stress analysis for predicting interface failure near crack-tips, and (3) the lack of an integrated framework that couples cure-dependent material properties with a multiscale modeling approach—this study proposes a FEM framework tailored to FOPLP. This framework combines a global model for overall thermomechanical behavior with a detailed submodel capturing localized stresses near copper pillars. This hierarchical approach enhances accuracy in predicting delamination and warpage. The framework incorporates cure-dependent thermal expansion behavior, capturing coupled thermal and chemical effects. Results support optimization of key process variables, minimizing warpage and improving reliability and yield. The model also quantifies EMC chemical shrinkage at elevated temperatures and resolves internal stress distributions under thermal cycling.

These insights are critical for assessing failure risks and guiding material and structural design. By integrating experimental data with advanced numerical techniques such as the VCCT, the framework provides a predictive tool for managing residual stress, reducing delamination, and enhancing the mechanical robustness and cost-efficiency of FOPLP in next-generation consumer electronics.

## 2 Theoretical Background of Evaluating Fracture Toughness at the Polyimide–Glass Interface

During the detachment of a FOPLP from its glass carrier, complex internal stress fields arise due to the interplay of material properties and interfacial mechanics. The critical stress magnitude at crack initiation is governed by the interfacial adhesion strength. To quantify the adhesion between the PI layer and the glass substrate, a bimaterial fracture mechanics approach was adopted. Given the activation of mixed-mode failure mechanisms, a combined experimental–computational methodology was employed, integrating peeling tests with high-fidelity finite element simulations to elucidate the interfacial fracture behavior.

**2.1 Bimaterial Fracture Mechanics.** In bimaterial interfacial fracture analysis, crack-tip behavior typically involves a combination of opening (mode I) and sliding (mode II) deformations. Due to oscillatory stress singularities inherent to bimaterial interfaces, the individual strain energy release rate components,  $G_I$  and  $G_{II}$ , often exhibit divergence near the crack tip. However, their algebraic sum, the total strain energy release rate  $G_T$ , remains well-behaved and convergent [39]. Using Irwin’s crack closure integral approach, a theoretical relationship has been established between  $G_T$  and the mixed-mode stress intensity factors  $K_I$  and  $K_{II}$ , capturing the energy–stress interplay in bimaterial systems. This relationship is expressed as follows [40]:

$$G_T = \frac{1}{16} \left[ \frac{\kappa_1 + 1}{\mu_1} + \frac{\kappa_2 + 1}{\mu_2} \right] (K_I^2 + K_{II}^2) \quad (1)$$

$\mu_1$  and  $\mu_2$  are shear moduli of materials 1 and 2, respectively. The terms  $\kappa_1$  and  $\kappa_2$  are dimensionless material parameters corresponding to each material. The value of  $\kappa_j$  (for  $j = 1, 2$ ) is selected based on the assumed stress state, as defined below

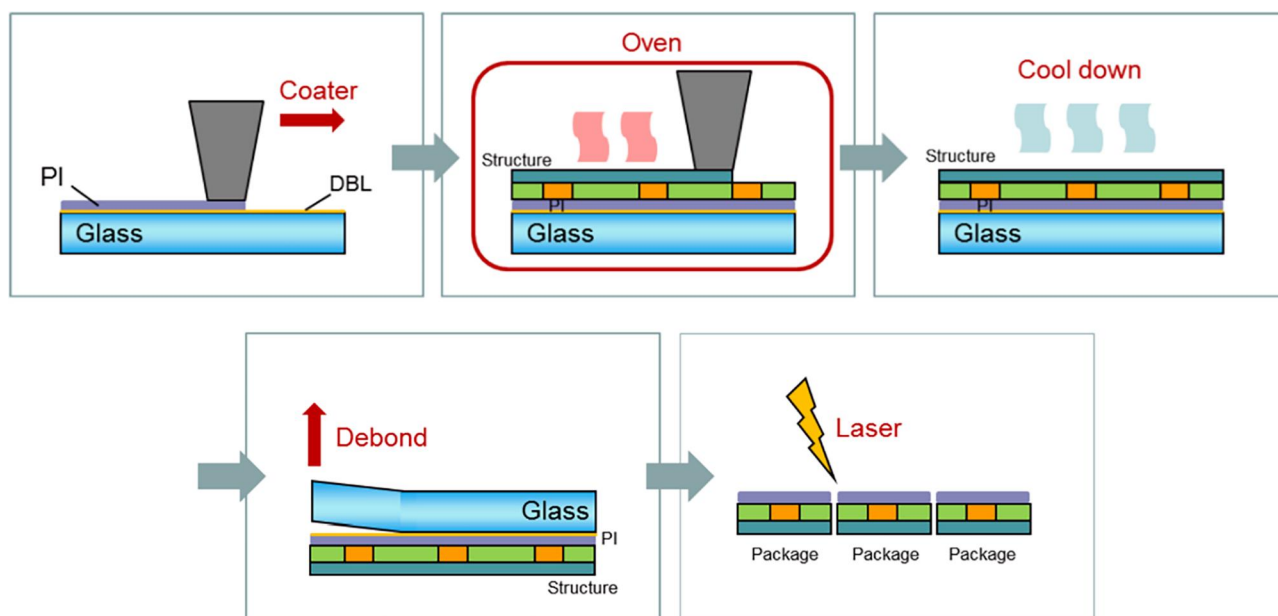


Fig. 1 Schematic of FOPLP detachment

$$\kappa_j = \begin{cases} 3 - 4\nu_j & \text{plane strain} \\ (3 - \nu_j)/(1 + \nu_j) & \text{plane stress} \end{cases} \quad (2)$$

$\nu_1$  and  $\nu_2$  are Poisson's ratios of materials 1 and 2, respectively. In FEM analysis, the total strain energy release rate  $G_T$  can be directly obtained using the modified crack closure integral method proposed by Ref. [31] as follows:

$$\kappa_j = \frac{1}{2\delta a} \left[ F_x^b (u_x^a - u_x^{a'}) + F_y^b (u_y^a - u_y^{a'}) \right] \quad (3)$$

$\delta a$  is the infinitesimal crack extension,  $F_x^b$  and  $F_y^b$  are the forces at the crack tip, and  $u_x^a$  and  $u_y^a$  are the displacement nearest to the crack tip, as illustrated in Fig. 2.

In bimaterial fracture, the mode mixity, associated with the stress intensity factors of modes I and II, as defined by the following equations:

$$\psi = \tan^{-1} \left( \frac{K_{II}}{K_I} \right) \quad (4)$$

$\frac{K_{II}}{K_I}$  can be further defined in terms of the displacement ratio as

$$\frac{K_{II}}{K_I} = \frac{H_1 - H_2 (\Delta u_y / \Delta u_x)}{H_2 + H_2 (\Delta u_y / \Delta u_x)} \quad (5)$$

$H_1$  and  $H_2$  are explicitly expressed as follows:

$$H_1 = \cos(R) + 2\varepsilon \sin(R) \quad (6)$$

$$H_1 = \sin(R) - 2\varepsilon \cos(R) \quad (7)$$

$$R = \varepsilon \ln \left( \frac{r}{l_k} \right) \quad (8)$$

and

$$\varepsilon = \frac{1}{2\pi} \ln \left( \frac{\kappa_1/\mu_1 + 1/\mu_2}{\kappa_2/\mu_2 + 1/\mu_1} \right) \quad (9)$$

$\Delta u_x$  and  $\Delta u_y$  in Eq. (5) are the relative displacements in the  $x$  and  $y$  directions, respectively. In Eq. (8),  $r$  represents the distance measured from the crack tip, and  $l_k$  is a characteristic length used for normalizing the distance  $r$  from the crack tip. Although theoretically arbitrary, its value is typically chosen to be a characteristic dimension of the problem, such as crack length or layer thickness. In finite element analysis, a common and convenient choice for  $l_k$  is the size of the elements at the crack tip. Building on the theoretical foundation described above, the interfacial properties between the PI layer and the glass substrate in FOPLP are examined, along with the stress distribution that develops during the

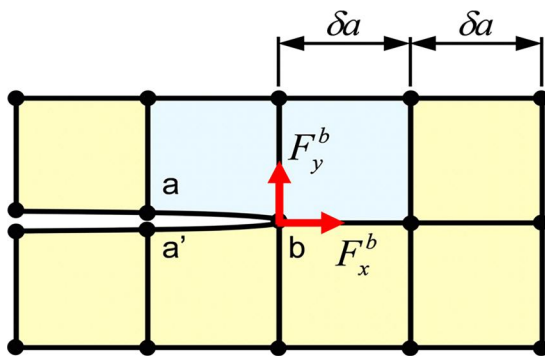


Fig. 2 Schematic of crack closure technique

delamination process as the panel is separated. Initially, the critical force required for delamination is determined through experimental observation of interfacial crack propagation. Subsequently, the fracture toughness and mode mixity of the interface are quantified using the crack closure integral method and the displacement ratio method, in conjunction with FEM analysis.

**2.2 Virtual Crack Closure Technique.** This study employs the VCCT to calculate the strain energy release rate along the crack front. The methodology is based on the crack closure integral approach originally developed by Shivakumar et al. [35], with subsequent refinements to enhance accuracy in three-dimensional fracture simulations. The technique assumes that the energy released during crack propagation over an incremental length  $\delta a$  is equivalent to the energy required to close the crack over the same distance. To satisfy this assumption, the mesh near the crack front must be orthogonally structured to ensure accurate numerical evaluation [34]. Given the limitations of the original formulation for nonorthogonal meshes, an established, modified VCCT expression is adopted to improve reliability in computing the strain energy release rate. The development of such modified techniques is well-documented, evolving from the foundational principles of the crack closure integral, to its extension for three-dimensional bodies, and subsequent refinements for handling general, skewed, and non-symmetric finite element meshes [34–36,41–46]. The relevant formulations are provided below

$$G_i^J = \frac{1}{2 \left[ S_1^J - \frac{1}{4} (S_1^J - S_2^J) \right]} (C^a v_i^a P_i^a + C^b v_i^b P_i^b), \quad i = 1, 2, 3 \quad (10)$$

The superscript  $J$  denotes the  $J$ th mesh element along the crack front, while the superscripts  $a$  and  $b$  represent the relative nodes within that element, as schematically illustrated in Fig. 3. While this notational concept is general, the specific implementation of the VCCT calculation, particularly the number of nodes involved and the values of the weighting constants ( $C$ ) in Eq. (11), is dependent on the type of mesh element used (e.g., linear four-node versus higher-order eight-node elements).

$$C^a = \frac{S_1^J}{S_1^{J-1} + S_1^J}; \quad C^b = \frac{S_1^J}{S_1^{J-1} + S_1^J} \quad (11)$$

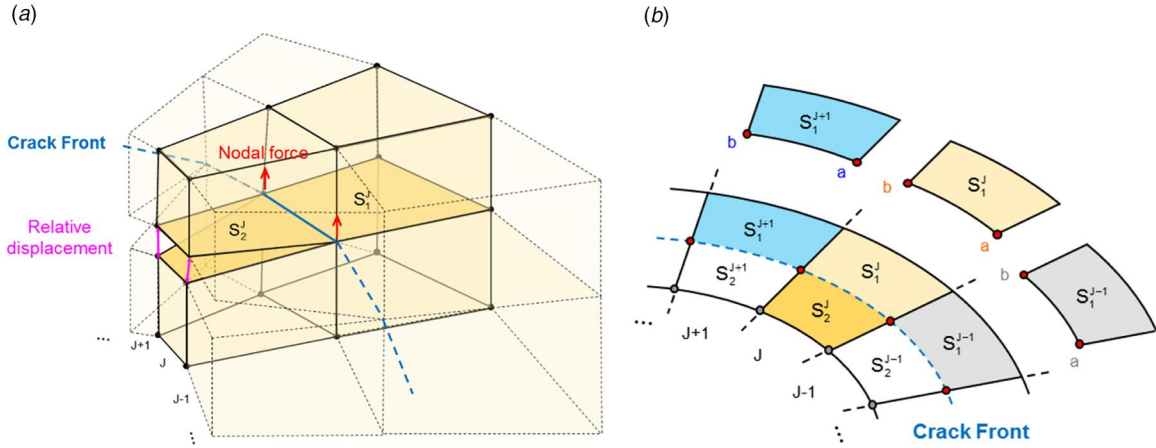
The total strain energy release rate is determined as the summation of the strain energy release rates corresponding to each failure mode, as expressed in following equation:

$$G_T = G_1 + G_2 + G_3 \quad (12)$$

### 3 Mechanical Debonding Process of Fan-Out Panel-Level Packaging

Chip delamination during FOPLP assembly may result from excessive warpage-induced stress or mechanical separation. To mitigate failure risk and reduce experimental cost and development time, FEM was employed to analyze warpage and internal stress distribution, with the goal of minimizing structural deformation and preventing chip detachment. This study examines mechanical debonding and thermally induced warpage in panel-level packaging. The panel consists of an array of  $25 \times 25 \text{ mm}^2$  modules, each embedding two  $10 \times 10 \text{ mm}^2$  semiconductor dies. A three-module model was developed to simulate debonding along the negative  $x$ -axis. The experimental setup and multilayer cross section are shown in Figs. 4 and 5. To balance model fidelity and computational efficiency, a hierarchical FEM strategy was adopted, combining a global model for overall deformation with a localized submodel to resolve critical stress concentrations.

**3.1 Construction of Effective Global Model.** In simulating the debonding process in FOPLP, three encapsulated modules were

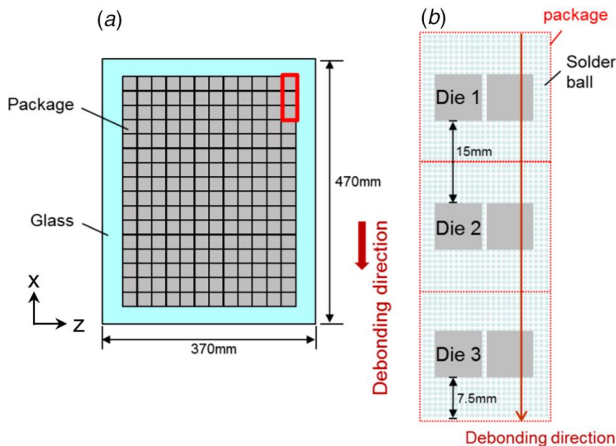


**Fig. 3 Schematic of the VCCT: (a) crack tip in a three-dimensional eight-noded element and (b) detailed view of the three-dimensional crack plane**

selected as the global model for structural analysis. The geometric configuration of this model is shown in Fig. 6. Each material layer was systematically simplified and assigned effective material properties to establish a computationally efficient global representation. The modeling methodology is outlined in Fig. 7. The structure comprises, from bottom to top: EMC, embedded die, aluminum pad, PI, silicon nitride (SiN<sub>x</sub>), under bump metallurgy (UBM), copper pillar, SnAg solder, underfill, three RDLs, and glass. Within the RDLs, horizontally distributed copper traces and their associated dielectric layers are represented as RDL1, RDL2, and RDL3, while the vertically interconnected regions are designated as RDL1\_D1, RDL2\_D1, and RDL3\_D1. Geometric simplification and material homogenization were applied to each layer to construct the effective global model, as shown in Figs. 7(b) and 7(c). The material properties used in the simulation, including experimentally derived effective values (indicated by \*), are summarized in Table 1. For the copper pillar region, an effective material model was also implemented. The simplified structure yielded a volume fraction of 8.8% for copper pillars ( $f_s$ ) and 91.2% for filler material ( $f_{ii}$ ). The equivalent structural configuration is schematically illustrated in Fig. 8.

The analytical method used here is based on the homogenization concept from composite mechanics [47]. Using this approach, the effective in-plane and out-of-plane CTEs, denoted by  $\alpha_{x,y}$  and  $\alpha_z$ , respectively, are calculated via the following equations:

$$\alpha_z = \frac{E_1 \alpha_1 \xi_1 + E_2 \alpha_2 \xi_2}{E_1 \xi_1 + E_2 \xi_2} \quad (13)$$



**Fig. 4 Schematic of the FOPLP: (a) top view and (b) enlarged in of top view**

$$\alpha_{x,y} = (1 + \nu_1) \alpha_1 \xi_1 + (1 + \nu_2) \alpha_2 \xi_2 - \alpha_z \bar{\nu} \quad (14)$$

where  $E_1(E_2)$ ,  $\alpha_1(\alpha_2)$ , and  $\xi_1(\xi_2)$  are Young's modulus, CTE, and volume fraction of the copper in RDL, respectively. In Eq. (14), the effective Poisson's ratio  $\bar{\nu}$  can be assessed by using rule of mixture,

$$\bar{\nu} = \nu_1 \xi_1 + \nu_2 \xi_2 \quad (15)$$

$\nu_1$  and  $\nu_2$  are the Poisson's ratio of the Cu and dielectric layer, respectively. Similarly, the effective in-plane elastic modulus  $E_{x,y}$  and out-of-plane elastic modulus  $E_z$  of the RDLs are calculated as

$$E_{x,y} = \frac{E_1 E_2}{E_1 \xi_2 + E_2 \xi_1} \quad (16)$$

$$E_z = E_1 \xi_1 + E_2 \xi_2 \quad (17)$$

The effective mechanical properties derived from this approach are summarized in Table 2.

The simulation in this study is conducted under the plane strain assumption to accurately represent the stress state at the crack front. In this framework, the analysis is performed on the x-y plane (the model's cross section), with the assumption that the strains in the out-of-plane z-direction are negligible ( $\epsilon_{zz} = \epsilon_{xz} = \epsilon_{yz} = 0$ ). This approach is justified because the FOPLP structure is very long in the z-direction relative to its thickness, causing the interior regions to be highly constrained. Although the overall model is three-dimensional, this constraint ensures the local stress field near the long, straight crack front behaves under plane strain conditions, which is a valid and standard approximation in three-dimensional fracture mechanics. To capture the stress concentration at the crack tip, a refined and uniformly structured mesh is implemented in the vicinity of the crack, with a minimum element size of 0.001 mm. The finite element model consists of 101,744 elements, as shown in Fig. 9. Boundary conditions are applied to replicate the mechanical debonding process. Specifically, the rightmost edge ( $x = 75$  mm), where stress influence from delamination is minimal, is fully constrained to eliminate rigid body motion and improve numerical convergence. Additionally, a uniform downward pressure of  $-92$  kPa is applied to the bottom surface to replicate the vacuum suction conditions observed during experimental testing.

Upon construction of the effective global model, the debonding simulation process is carried out, as illustrated in Fig. 10. The interfacial fracture toughness is set to  $G_C = 6$  J/m<sup>2</sup>, based on experimental measurements. A detailed stress analysis is then performed across all material layers, and the results are presented in Table 3. An asterisk (\*) denotes brittle materials, where the reported values correspond to the maximum principal stress observed within each layer.

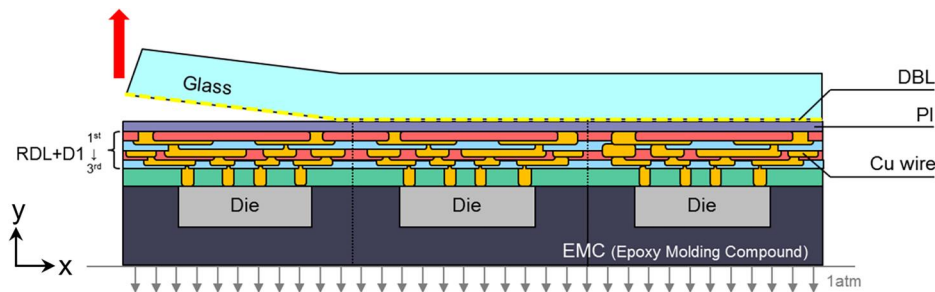


Fig. 5 Schematic of local side view of FOPLP

The simulation results reveal that, aside from the glass and the first PI layer, the SiNx\_1 layer exhibits the highest stress within the effective global model. It is acknowledged that the predicted stress distribution is approximate due to the structural simplifications and effective material approximations employed in the model. As a result, localized stress gradients within the packaging structure may not be fully resolved. Notably, the global model indicates pronounced stress concentration near the crack tip. To enable a more detailed assessment of the stress behavior in this critical

region, a refined submodel is developed, specifically targeting the crack tip zone for localized analysis.

**3.2 Stress Analysis of the Copper Pillar Sub-Model.** Section 3.1 described the finite element simulation framework for FOPLP, emphasizing the importance of integrating both global and submodels. Consistent with the global analysis, the submodel was constructed as a two-dimensional domain under the plane strain

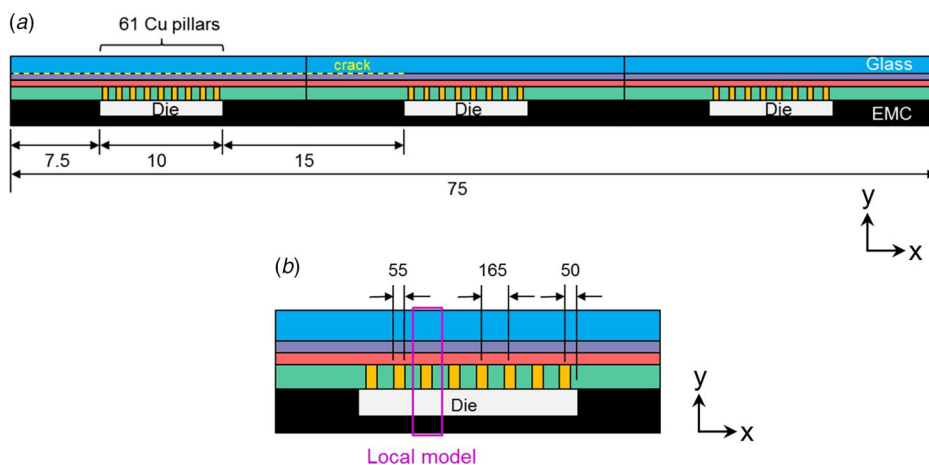


Fig. 6 Schematic of global model: (a) configuration of three encapsulated packages and (b) exhibition of a unit package

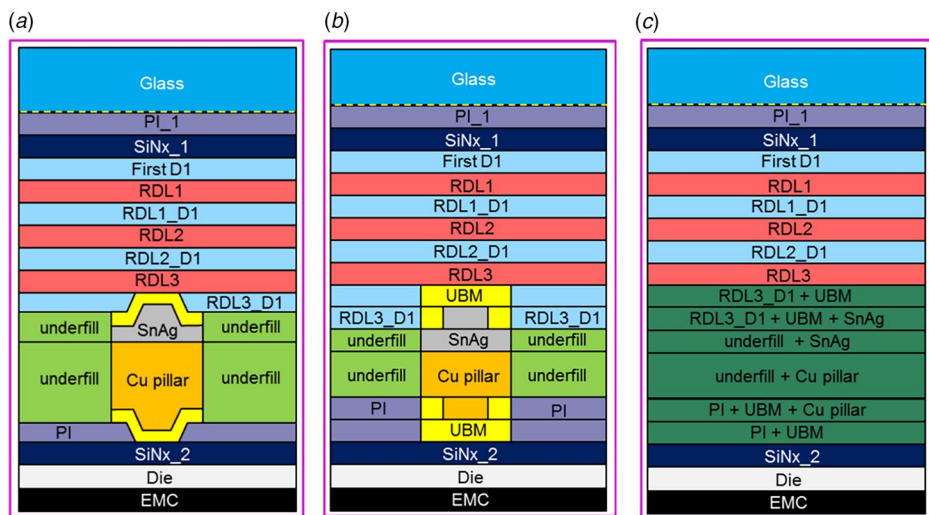
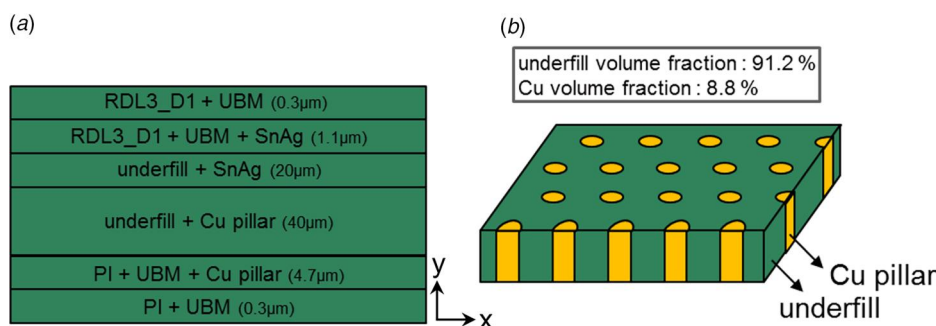


Fig. 7 Schematic of the local structure in the FOPLP model: (a) side view, (b) side view after geometric simplification, and (c) side view of effective model

**Table 1 Mechanical properties of each material layer in the FOPLP structure**

| Material       | Thickness ( $\mu\text{m}$ ) | Young's modulus (GPa) | Poisson's ratio | CTE ( $10^{-6}/^\circ\text{C}$ ) |
|----------------|-----------------------------|-----------------------|-----------------|----------------------------------|
| Glass          | 700                         | 90.95                 | 0.3             | 3.8                              |
| PI_1           | 15                          | 6.5                   | 0.3             | 3.76                             |
| SiNx           | 1                           | 112                   | 0.24            | 2.9                              |
| First D1       | 1                           | 36                    | 0.22            | 13                               |
| *RDL_1         | 3.6                         | 52.8                  | 0.24            | 18.14                            |
| RDL1_D1        | 1.4                         | 36                    | 0.22            | 56.0                             |
| *RDL_2         | 3.6                         | 52.8                  | 0.07            | 18.14                            |
| RDL2_D1        | 1.4                         | 36                    | 0.22            | 56.0                             |
| *RDL_3         | 3.6                         | 52.8                  | 0.07            | 18.14                            |
| RDL3_D1        | 1.4                         | 36                    | 0.22            | 56.0                             |
| *UBM           | 0.3                         | 114.2                 | 0.33            | 15.3                             |
| SnAg           | 20                          | 55                    | 0.365           | 21.0                             |
| Cu pillar      | 40                          | 120                   | 0.4             | 17.5                             |
| Underfill      | 60                          | 11                    | 0.36            | 25.0                             |
| PI_2           | 5                           | 6.5                   | 0.3             | 3.76                             |
| *Al Pad + SiNx | 2.8                         | 197.5                 | 0.24            | 5.02                             |
| Die            | 250                         | 131                   | 0.26            | 2.8                              |
| EMC            | 100                         | 21.5                  | 0.3             | 9 ( $<T_g$ )<br>45 ( $>T_g$ )    |

**Fig. 8 The modeling approach for simplifying complex multilayer structures: (a) schematic of effective layers and (b) schematic of the effective structure**

assumption. While the global model was detailed previously, this section focuses on the construction of the submodel and the corresponding localized stress analysis. The submodel targets the region exhibiting the highest stress concentration, as identified near the crack tip in the global simulation. A domain encompassing a single copper pillar was selected, with the pillar centrally positioned to capture stress localization accurately. The submodel dimensions are  $449.8 \mu\text{m} \times 165 \mu\text{m}$ . The chamfer geometry of the upper and lower UBM regions was defined based on scanning electron microscope observations (Fig. 11), with the corresponding configuration shown in Fig. 12.

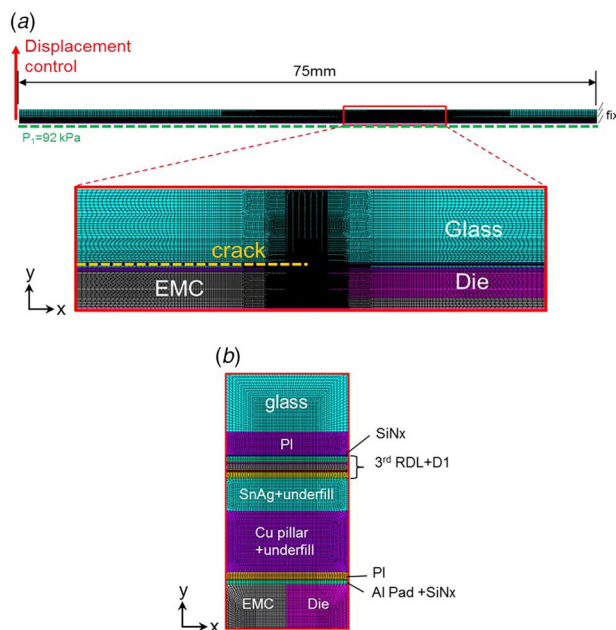
To resolve local stress gradients, a refined mesh was applied near the copper pillar, with a minimum element size of  $0.05 \mu\text{m}$ . The total element count was 76,718 (Fig. 13). Displacement data from the global model at the onset of interfacial failure were interpolated and imposed on the submodel boundaries. This displacement-mapping approach ensures mechanical consistency between models and enables accurate evaluation of the interfacial stress field through localized finite element analysis.

**3.3 Analysis of Crack Propagation Locations.** During the debonding process of panel-level packaging structures, excessive stress concentrations can lead to crack propagation and compromise structural integrity. As stress distribution varies with the crack propagation path and its proximity to copper pillars, it is essential to identify critical crack locations associated with maximum stress to guide design optimization and failure mitigation. This study investigates the relationship between crack propagation positions and local stress concentrations in copper pillars. Six distinct crack locations were evaluated in the global finite element model, designated as points A through F: point A is at the rear end of the first die; point B at the front end of the second die; point C directly above a copper pillar in the front-middle of the second die; point D at the die center; point E above a copper pillar in the rear-middle; and point F at the rear end of the second die. These locations are shown in Fig. 14.

For each crack propagation case, the copper pillar nearest to the crack tip was selected for detailed analysis. To accommodate geometric differences and spatial positioning, three submodels were

**Table 2 Mechanical properties of effective layers in the FOPLP structure**

| Material              | Thickness ( $\mu\text{m}$ ) | $E_{\text{eff},y}$ (GPa) | $E_{\text{eff},x}, E_{\text{eff},z}$ (GPa) | $\text{CTE}_{\text{eff},y}$ ( $10^{-6}/^\circ\text{C}$ ) | $\text{CTE}_{\text{eff},x}, \text{CTE}_{\text{eff},z}$ ( $10^{-6}/^\circ\text{C}$ ) | $\nu_{\text{eff}}$ |
|-----------------------|-----------------------------|--------------------------|--------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------|
| RDL3_D1 + UBM         | 0.3                         | 42.9                     | 38.4                                       | 4.01                                                     | 1.85                                                                                | 0.23               |
| RDL3_D1 + UBM + SnAg  | 1.1                         | 37.4                     | 37.2                                       | 3.19                                                     | 2.36                                                                                | 0.23               |
| underfill + SnAg      | 20                          | 14.9                     | 12.2                                       | 23.52                                                    | 24.65                                                                               | 0.31               |
| underfill + Cu pillar | 40                          | 20.6                     | 12.6                                       | 20.78                                                    | 24.34                                                                               | 0.36               |
| PI + UBM + Cu pillar  | 4.7                         | 16.5                     | 7.5                                        | 15.52                                                    | 12.48                                                                               | 0.31               |
| PI + UBM              | 0.3                         | 16.0                     | 7.5                                        | 14.06                                                    | 12.29                                                                               | 0.30               |



**Fig. 9 FEM model and boundary conditions: (a) global FEM model and (b) FEM model at the crack tip**

constructed (Fig. 15). In cases A and F, the crack tip extends beyond the copper pillar. In cases C, D, and E, the tip is positioned directly above the pillar, while in case B it lies just before it. Consistent meshing was applied across all submodels to ensure numerical comparability. The maximum stress values from the nearest copper pillar in each case are summarized in Table 4.

The analysis reveals that the highest stress occurs when the crack propagates to the front end of the die (point B), where mechanical loading is typically concentrated. Middle-die regions exhibit relatively stable and lower stress values, while the rear end (point F) shows the lowest stress due to reduced loading. In all cases, the maximum stress is localized at the lower right region of the copper pillar, specifically at the SiNx<sub>2</sub>–UBM interface corner (Fig. 16), indicating a critical site for failure initiation. Based on these findings, subsequent parametric analyses are standardized using point B as the crack location, with a focus on stress behavior in the UBM layer due to its significance in maintaining packaging structural reliability.

**3.4 Parametric Study of Debonding Stress in the Submodel.** Based on the crack propagation analysis, the condition corresponding to propagation at point B in the global model is selected as the reference case for subsequent parametric studies of debonding stress. The following parameters were chosen because they represent the most direct geometric and material factors influencing the stress concentration at the critical SiNx–UBM interface. Three key parameters are investigated to reduce stress across various material layers. These include the geometry of UBM etching chamfers, the interfacial fracture toughness of the debonding layer, and the width of UBM etching openings.

The effect of UBM etching chamfers is first analyzed, as stress concentrations primarily occur at the interface between the UBM and the copper pillar due to abrupt geometric transitions. During fabrication, an etched groove is formed, followed by the deposition

**Table 3 Maximum stress in each layer of the effective global model for the FOPLP**

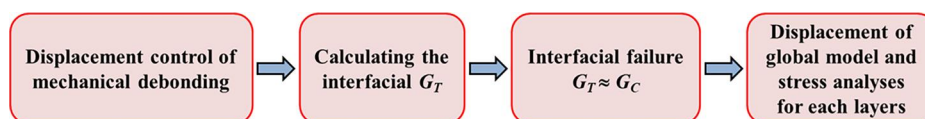
| Material         | Maximum von-Mises stress (MPa) | Maximum principal stress (MPa) |
|------------------|--------------------------------|--------------------------------|
| Glass            | 150.5                          | —                              |
| PI_1             | 123.3                          | —                              |
| SiNx_1*          | —                              | 67.6                           |
| First D1         | 33.1                           | —                              |
| RDL_1            | 32.9                           | —                              |
| RDL1_D1          | 33.1                           | —                              |
| RDL_2            | 37.8                           | —                              |
| RDL2_D1          | 32.8                           | —                              |
| RDL_3            | 40.1                           | —                              |
| RDL3_D1          | 33.2                           | —                              |
| UBM              | 33.0                           | —                              |
| SnAg             | 25.4                           | —                              |
| Cu pillar        | 24.7                           | —                              |
| Underfill        | 25.6                           | —                              |
| PI_2             | 25.7                           | —                              |
| Al Pad + SiNx_2* | —                              | 56.2                           |
| Die*             | —                              | 57.5                           |
| EMC              | 23.2                           | —                              |

of the UBM layer that encapsulates the copper pillar. Scanning electron microscope observations confirm the existence of chamfers at the etched corners. To account for this feature, chamfers are incorporated at both upper and lower UBM corners in the submodel. The impact of chamfer geometry on stress distribution is illustrated in Fig. 17.

Three submodels with different chamfer radii are constructed, namely, submodels I, II, and III. The corresponding upper and lower chamfer radii, R1 and R2, are listed in Table 5. As detailed in Sec. 3.3, stress concentrations are prominent at the lower right corner of SiNx<sub>2</sub> and UBM<sub>2</sub>, as well as the upper left corner of UBM<sub>1</sub>. These regions are analyzed while excluding distorted mesh elements near chamfered corners to avoid numerical artifacts. As shown in Table 6, increasing the chamfer radius significantly mitigates local stress concentrations, consistent with established mechanical principles. However, due to fabrication constraints, precise control over chamfer radius remains challenging, and implementation in manufacturing requires further process refinement.

The influence of the interfacial fracture toughness of the debonding layer is also examined. A lower fracture toughness facilitates easier release during debonding and reduces stress transmitted to adjacent structural components. By adjusting the hydrogen bond concentration, the fracture toughness is varied from 1.5 to 8 J/m<sup>2</sup>. This range represents realistic values for typical adhesive systems. Finite element simulations are performed to evaluate stress levels in the UBM layer, which is more susceptible to damage. The results, summarized in Table 7, indicate that increasing fracture toughness leads to a corresponding rise in von Mises stress in UBM<sub>2</sub>. This behavior aligns with the expectation that greater resistance to crack growth results in higher localized stresses. Incorporating a debonding layer with reduced fracture toughness can therefore reduce the risk of structural failure.

Lastly, the impact of UBM etching opening geometry on stress distribution is investigated. In the simulation, the copper pillar diameter remains fixed at 55 mm, while the upper and lower etching opening widths are varied. These parameters are denoted as opening B and opening A, respectively. The etching angle at the UBM<sub>2</sub> corner, defined as  $\theta$ , is also shown in Fig. 17. Four submodels are



**Fig. 10 Flowchart of the debonding process analysis for the FOPLP**

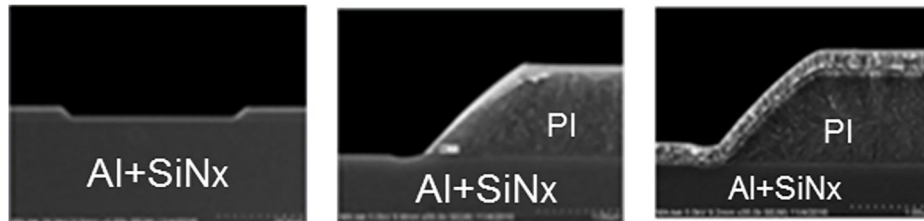


Fig. 11 SEM images of copper pillar chamfering in the UBM etching process

constructed with opening angles of 116.6 deg, 135 deg, 158.2 deg, and 161.6 deg. The corresponding opening dimensions are provided in Table 8. Stress analysis is performed on surrounding layers including RDL3\_D1, UBM\_2, PI\_2, and SiNx\_2. The von Mises stress is evaluated for the ductile metallic and polymeric layers (UBM, RDL, PI), while the maximum principal stress is used for the brittle ceramic layer (SiNx) to apply the most appropriate failure criterion for each material. As shown in Table 9, an increase in etching angle results in lower stress in UBM\_2 and SiNx\_2, while PI\_2 and RDL3\_D1 experience a relative increase in stress. This redistribution reflects how geometric changes affect load transfer among materials. Given the susceptibility of UBM\_2 to delamination, the configuration with the largest opening angle of 161.6 deg is identified as the most favorable for minimizing critical stress and enhancing structural reliability.

#### 4 Debonding Analysis of Fan-Out Panel-Level Packaging Considering Process-Induced Residual Stresses

In the development of FOPLP, a simplified packaging structure was introduced to facilitate process testing and reduce experimental time and cost. It is important to note for the reader that the coordinate system convention in this section differs from that used in Sec. 3. For the following 3D thermal warpage and stress analyses, the panel lies on the  $x$ - $y$  plane, and the  $z$ -axis represents the through-thickness direction, consistent with standard plate and shell modeling conventions. Therefore, the out-of-plane stresses relevant to delamination are those in the  $z$ -direction. Compared to the full-scale structure, the simplified version features a smaller footprint

and a reduced number of material layers. From bottom to top, the stack comprises A1 glass, PI, embedded die, and EMC. To meet different experimental requirements, two configurations were developed: one with a PI layer (Simplified Packaging Structure I) and one without (Simplified Packaging Structure II). The geometries of both configurations are illustrated in Fig. 18.

In Simplified Packaging Structure I, the edge spacing between the die and the package boundary is fixed at 3 mm, while the spacing between adjacent dies is adjustable to modify the overall die occupation ratio. The thickness of each layer is specified in Fig. 19. Additionally, a release agent is applied at the interface between the glass and PI layers to facilitate debonding, enabling clean separation during the release process.

**4.1 Manufacturing Process Simulation of Fan-Out Panel-Level Packaging Structures.** Fan-out panel-level packaging Structure I is selected for experimental validation due to its representative four-layer design, which supports reliability assessment under thermal cycling. The stack sequence consists of glass A1 at the bottom, followed by the PI layer, the die, and the EMC on top. The fabrication process begins with PI lamination onto a  $100 \times 100 \text{ mm}^2$  glass substrate at room temperature ( $25^\circ\text{C}$ ), followed by curing at  $450^\circ\text{C}$ . After cooling to room temperature, dies are mounted onto the cured PI layer. The assembly is then heated to  $150^\circ\text{C}$  for EMC encapsulation and bonding, followed by a final cooling stage back to  $25^\circ\text{C}$ . The process flow and associated temperature profile are illustrated in Figs. 20 and 21.

The mechanical properties of the constituent materials used in the model were provided by the material supplier and are summarized in

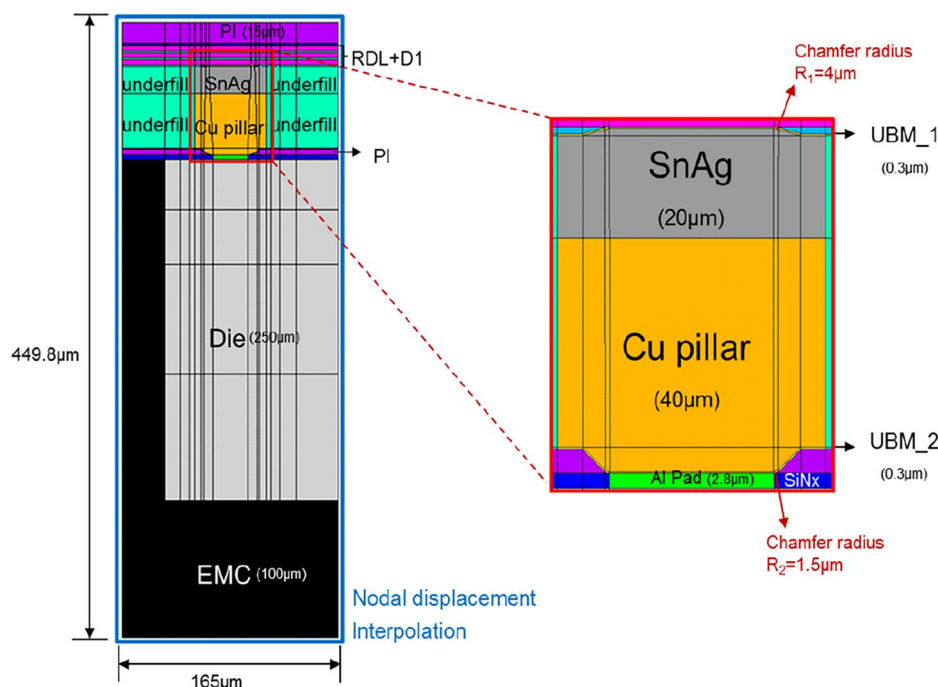


Fig. 12 Schematic of the copper pillar submodel

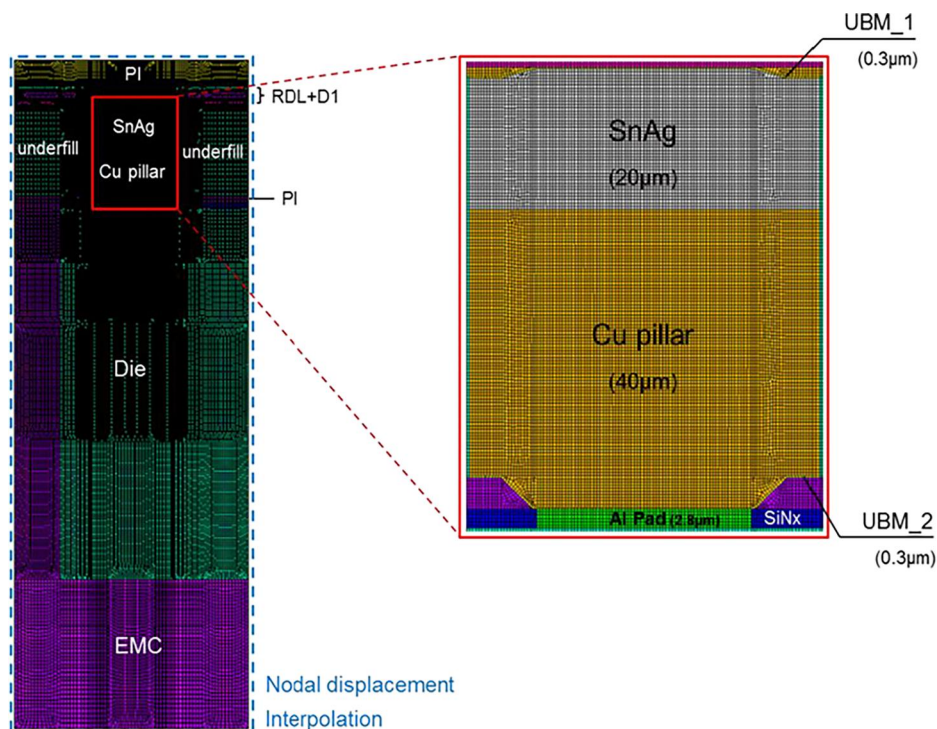


Fig. 13 FEM model of the copper pillar submodel

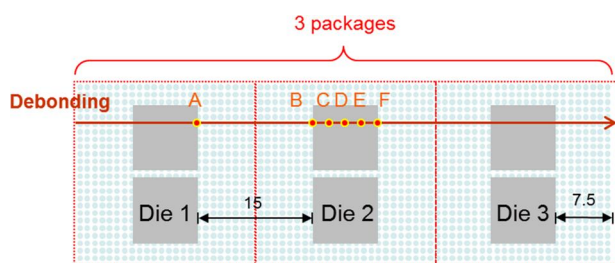


Fig. 14 Crack propagation locations in the global model

Table 4 Maximum stress in the copper pillar at the crack tip corresponding to different crack propagation locations

| Crack tip              | Point A | Point B | Point C | Point D | Point E | Point F |
|------------------------|---------|---------|---------|---------|---------|---------|
| Von-Mises stress (MPa) | 162.8   | 281.2   | 201.2   | 200.7   | 201.0   | 161.3   |

Table 10. Notably, the EMC is a thermosetting polymer that undergoes irreversible volumetric shrinkage due to internal chemical reactions at elevated temperatures. Based on the methodology for determining the chemical shrinkage coefficient

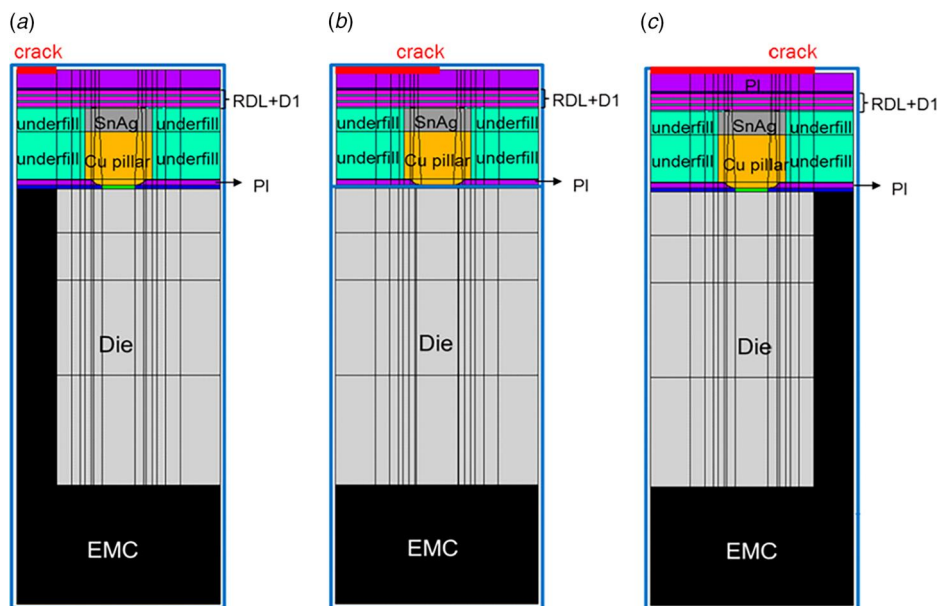


Fig. 15 Three submodels corresponding to different crack propagation locations: (a) Submodel at point B, (b) Submodels at points C, D, and E, and (c) Submodels at points A and F

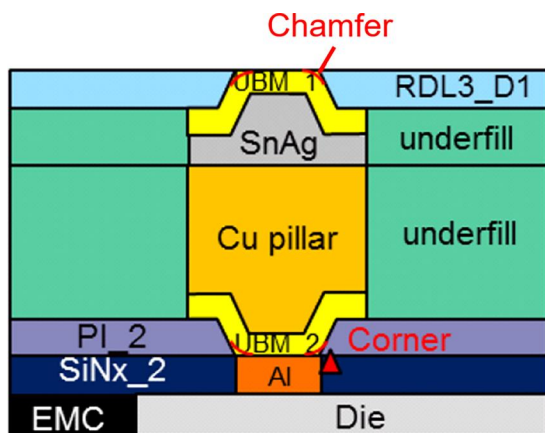


Fig. 16 Schematic of stress concentration regions near the copper pillar

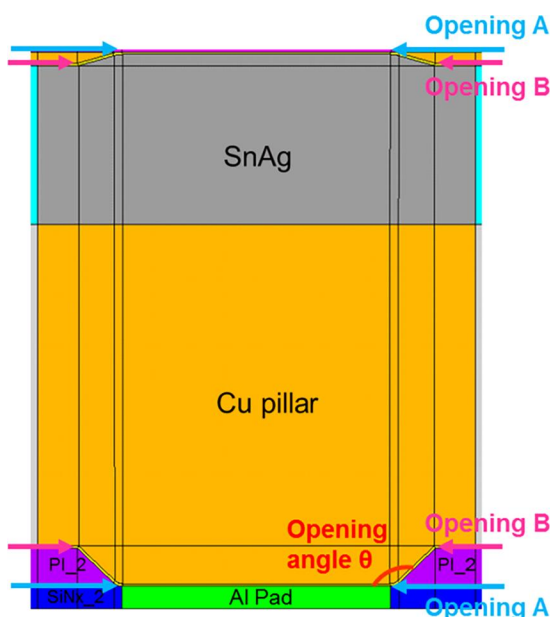


Fig. 17 Schematic of the model for upper and lower UBM etching openings

established in the authors' previous work [15], the chemical shrinkage coefficient of the EMC used in this study is identified as 0.96%. The relationship between EMC volumetric shrinkage and temperature is illustrated in Fig. 22. Furthermore, the Young's

Table 5 Etching chamfer radii corresponding to each submodel

| Chamfer radius       | Sub-Model I | Sub-Model II | Sub-Model III |
|----------------------|-------------|--------------|---------------|
| R1 ( $\mu\text{m}$ ) | 0           | 4            | 10            |
| R2 ( $\mu\text{m}$ ) | 0           | 1.5          | 10            |

Table 6 Maximum stress in the UBM for submodels with different etching chamfer radii

| von Mises stress(MPa) |             |              |               |
|-----------------------|-------------|--------------|---------------|
|                       | Sub-Model I | Sub-Model II | Sub-Model III |
| UBM_1                 | 137         | 133          | 103           |
| UBM_2                 | 197         | 176          | 112           |

Table 7 Maximum Stress in the UBM during debonding process for different interfacial fracture toughness values

| Fracture toughness $G_C$ (J/m <sup>2</sup> ) | von Mises stress (MPa) |
|----------------------------------------------|------------------------|
| 7.99                                         | 379.8                  |
| 6.05                                         | 329.5                  |
| 4.25                                         | 274.3                  |
| 2.96                                         | 226.8                  |
| 1.44                                         | 155.6                  |

modulus of EMC exhibits a distinct change across its  $T_g$  of 148 °C. The temperature-dependent variation in Young's modulus is shown in Fig. 23. A simplified, bilinear elastic model was adopted because it effectively captures the first-order effect of the large stiffness change across the  $T_g$  and has been validated against experimental warpage data (Sec. 4.2), while maintaining computational efficiency for this large-scale analysis.

**4.2 Warpage Simulation of the Fan-Out Panel-Level Packaging Structure.** In the warpage simulation of simplified packaging structure I, the initial die coverage was configured as  $10 \times 10$  array, and the final process temperature was set to room temperature (25 °C). However, experimental results revealed delamination in Simplified Packaging Structure I. To validate the accuracy of the simulation, four experimental trials were conducted under identical conditions using Simplified Packaging Structure II, which excludes the PI layer. Due to the absence of the PI layer, delamination did not occur after processing, allowing direct measurement of postprocess warpage for comparison with simulation results. The same simulation model was subsequently applied to investigate related issues in greater detail.

Warpage measurements for FOPLP structure II are presented in Table 11. The center-supported measurement method, depicted in Fig. 24, was employed, wherein the structure was placed on a granite surface and a feeler gauge was used to assess vertical displacement. However, achieving consistent central contact and accurate edge measurements presented challenges, potentially introducing variability into the warpage data. Based on the structural configuration illustrated in Fig. 18(b), a finite element model for FOPLP Structure II was developed. The EMC exhibits significant chemical shrinkage, concentrated in a narrow temperature interval near the end of the process. This shrinkage was implemented in the simulation over the temperature range from 149.8 °C to 150 °C. Given the temperature-

Table 8 Upper and lower opening widths for etching opening angles

| Opening angle $\theta$ (deg) | 116.6 | 135 | 158.2 | 161.6 |
|------------------------------|-------|-----|-------|-------|
| Opening A ( $\mu\text{m}$ )  | 35    | 35  | 20    | 20    |
| Opening B ( $\mu\text{m}$ )  | 40    | 45  | 45    | 50    |

Table 9 Maximum stress as a function of the etching opening angle

| von Mises stress(MPa)        |                      |                    |                      |                      |
|------------------------------|----------------------|--------------------|----------------------|----------------------|
|                              | $\theta = 116.6$ deg | $\theta = 135$ deg | $\theta = 158.2$ deg | $\theta = 161.6$ deg |
| RDL3_D1                      | 121.5                | 121.5              | 186.1                | 226.3                |
| UBM_2                        | 365.2                | 329.5              | 254.1                | 241.2                |
| PI_2                         | 176.0                | 198.5              | 196.9                | 204.4                |
| First principal stress (Mpa) |                      |                    |                      |                      |
|                              | $\theta = 116.6$ deg | $\theta = 135$ deg | $\theta = 158.2$ deg | $\theta = 161.6$ deg |
| SiNx_2                       | 574.6                | 556.3              | 390.5                | 368.3                |

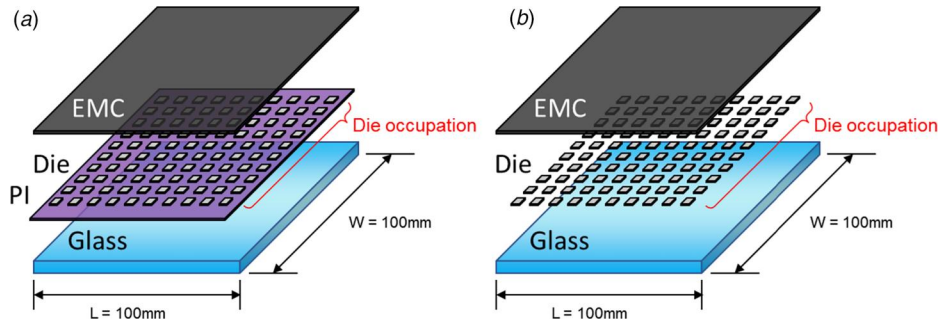


Fig. 18 The schematic of the two package configurations: (a) FOPLP structure I and (b) FOPLP structure II

dependent behavior of EMC, a bilinear kinematic hardening model was adopted to capture its plasticity beyond the glass transition temperature, as illustrated in Fig. 25. Due to the minimal yield strain assumed, the influence of the elastic region was considered negligible. All other materials were modeled as linearly elastic.

To optimize computational efficiency, a quarter-symmetry model consisting of 90,000 elements is employed. The boundary conditions and mesh configuration are depicted in Fig. 26. Displacements are constrained along the symmetry planes in the x- and y-directions, while the z-direction displacement is fixed at the origin to prevent rigid body motion. Warpage is defined as the difference in vertical displacement between the center and corner nodes at the bottom surface of the specimen. The simulation includes gravitational effects, and the resulting warpage, illustrated in Fig. 27, is 1.43 mm. This value falls within the experimental range of 1.3 mm–1.5 mm reported in Table 11, thereby validating the accuracy of the simulation framework for subsequent analysis.

**4.3 Stress Analysis of the Fan-Out Panel-Level Packaging Structure.** During thermal cycling of FOPLP Structure I, experimental observations confirmed delamination at the interface between the PI layer and the glass substrate. This outcome, not anticipated by the original process design, prompted a focused simulation-based investigation of the delamination mechanism. Notably, structural delamination is not solely determined by the warpage magnitude but is primarily governed by the strain energy release rate at the interfacial defect site. Delamination is initiated when this release rate exceeds the critical interfacial fracture toughness, consistent with fracture mechanics principles. As direct observation of delamination during oven baking is not feasible, a

two-step simulation approach is adopted: an initial defect-free model identifies regions of maximum interfacial stress, followed by a localized defect model to investigate delamination initiation.

The initial analysis focuses on evaluating stress distribution along the PI–glass interface after thermal loading. This enables the identification of potential initiation sites for microdefects, forming the basis for a more detailed simulation. Three-dimensional finite element models were constructed using the geometric configuration of FOPLP Structure I (Fig. 18). Two models were developed with die coverage ratios of  $10 \times 10$  and  $16 \times 16$ . A maximum element size of 0.5 mm was used, with refined mesh elements of 0.0625 mm near critical corner regions to capture stress gradients accurately. The models contained 245,664 and 237,600 elements, respectively, as shown in Fig. 28.

Thermal process simulations were performed, and interfacial stresses were evaluated using a local coordinate system. Specifically, the maximum normal stress in the z-direction ( $\sigma_{zz}$ ), and the shear stresses in the yz ( $\sigma_{yz}$ ) and xz ( $\sigma_{xz}$ ) directions were extracted at the PI–glass interface. As summarized in Table 12, the  $\sigma_{zz}$  component was consistently dominant, irrespective of the die coverage ratio. This observation aligns with theoretical predictions

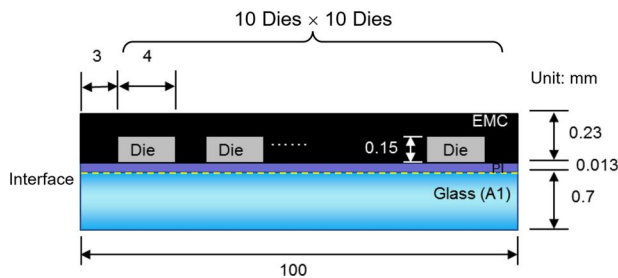


Fig. 19 Side view of FOPLP structure

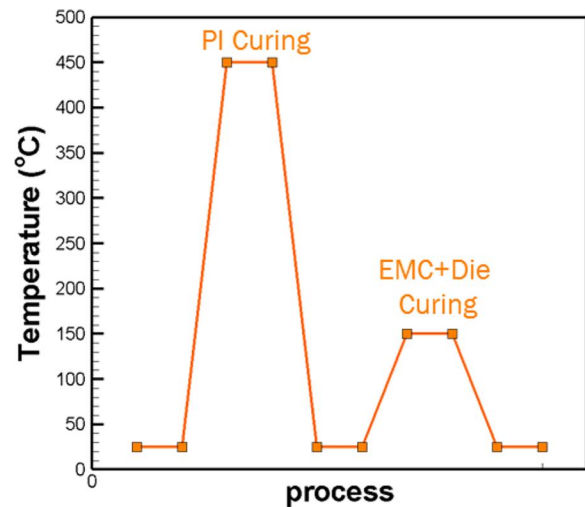


Fig. 21 Temperature profile for manufacturing process of FOPLP structure I

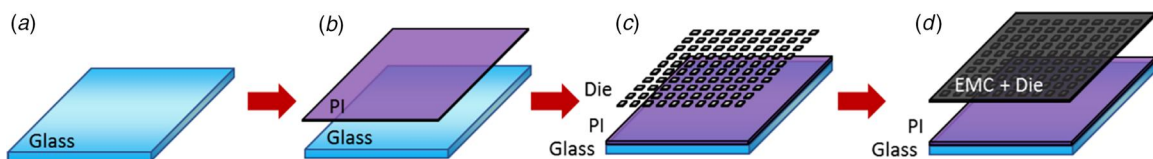
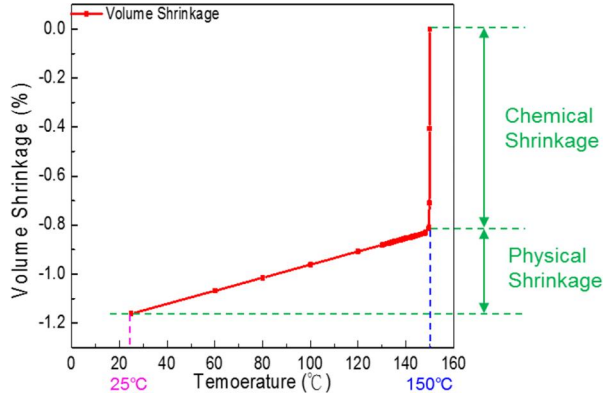


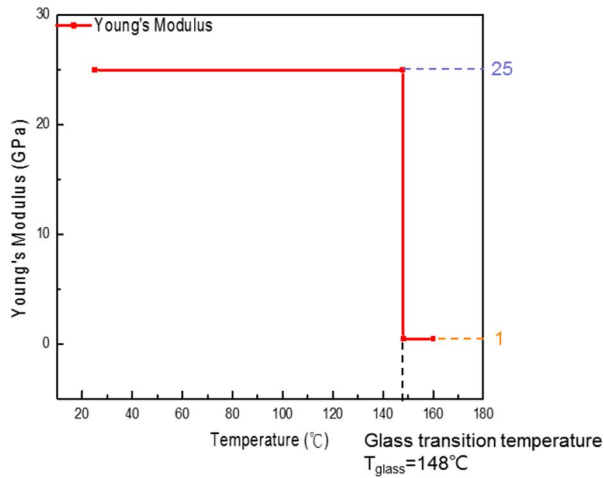
Fig. 20 Process flow of FOPLP structure I: (a) glass substrate, (b) PI coating, (c) die placement, and (d) EMC curing

**Table 10 Mechanical properties of materials in the simplified packaging structure**

| Material | Young's modulus (GPa)         | CTE                           | Poisson's ratio | Reference temperature ( $^{\circ}\text{C}$ ) |
|----------|-------------------------------|-------------------------------|-----------------|----------------------------------------------|
| EMC      | 25 ( $<T_g$ )<br>1 ( $>T_g$ ) | 9 ( $<T_g$ )<br>45 ( $>T_g$ ) | 0.3             | 150                                          |
| Die      | 131                           | 2.8                           | 0.26            | 150                                          |
| 45 U_Pi  | 5.2                           | 3.76                          | 0.4             | 450                                          |
| Glass A1 | 73.2                          | 3.8                           | 0.3             | 450                                          |



**Fig. 22 Volumetric shrinkage of EMC as a function of temperature**



**Fig. 23 Young's modulus of EMC as a function of temperature**

[48,49], as normal stresses typically drive delamination in multi-layer structures due to mismatches in material properties. Figure 29 presents the spatial distribution of  $\sigma_{zz}$ , revealing that peak stress consistently localizes at model corner point A. This location is identified as the most likely site for delamination initiation. Consequently, subsequent simulations introducing microdefects will focus on point A to evaluate crack growth behavior under thermal and mechanical loading.

**Table 11 Experimental measurement of warpage in packaging structure II**

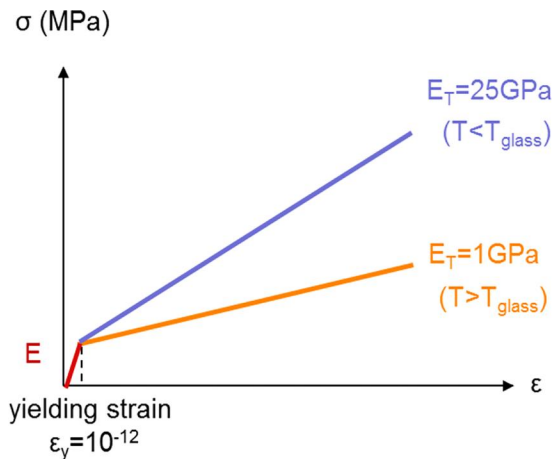
|              | Warpage (mm) |
|--------------|--------------|
| Experiment 1 | 1.30         |
| Experiment 2 | 1.25         |
| Experiment 3 | 1.30         |
| Experiment 4 | 1.50         |



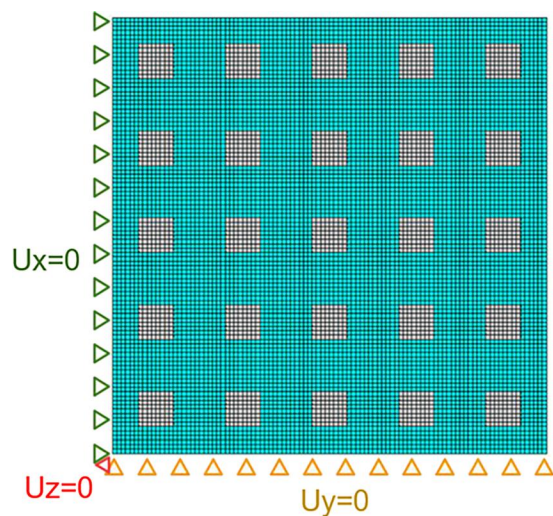
**Fig. 24 Warpage measurement using the central support method**

**4.4 Delamination Analysis of the Fan-Out Panel-Level Packaging Structure.** During the EMC curing and subsequent cooling stages, the entire FOPLP structure remained enclosed within an oven, rendering direct observation of delamination impractical. As a result, evaluations were limited to preliminary visual inspections intended to rapidly identify the initiation and extent of delamination. For FOPLP structure I with a  $10 \times 10$  die array, minor edge curling and signs of delamination were observed near  $70^{\circ}\text{C}$ . This temperature was therefore defined as the critical onset temperature for delamination. Based on the findings in Sec. 4.3, initial defects were assumed to originate at the package corners, as shown in Fig. 30. To assess structural integrity under thermal loading, FEM analysis was performed using a  $10 \times 10$  die layout of FOPLP structure I. Two configurations were simulated, with total element counts of 137,228 and 136,226, respectively. Symmetry boundary conditions were applied by constraining displacements in the x-direction on the x-symmetric plane, in the y-direction on the y-symmetric plane, and in the z-direction at the intersection of the three symmetry planes. These constraints ensured numerical stability and accurately represented physical symmetry. Details of the mesh and boundary conditions are shown in Fig. 31.

Experimental results confirmed that initial delamination consistently occurred near  $70^{\circ}\text{C}$ . To simulate this condition, a microdefect



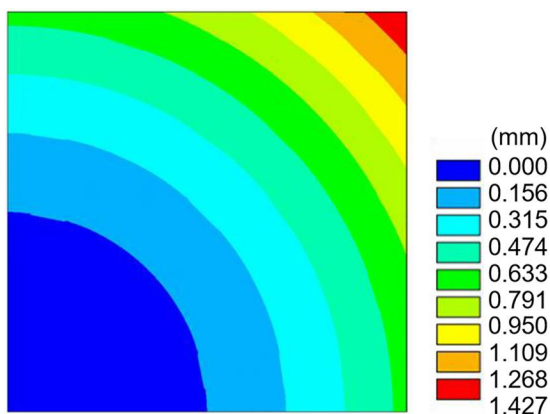
**Fig. 25 Young's modulus of EMC using a plastic model**



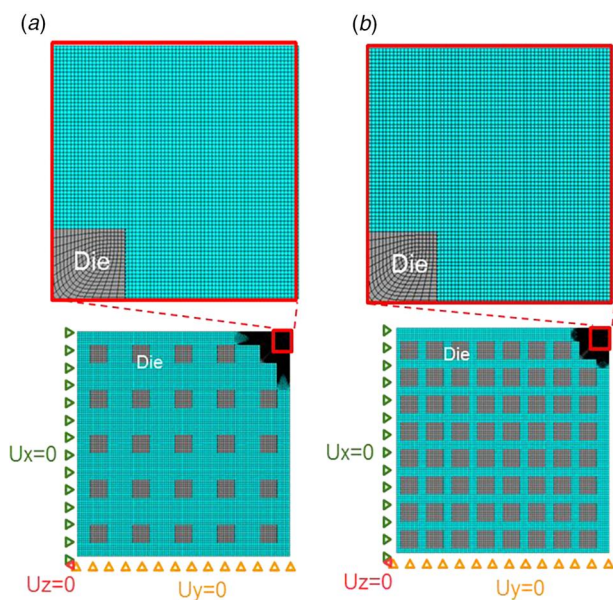
**Fig. 26** A quarter FEM model of FOPLP structure and boundary conditions

representing a thermally induced crack was introduced at the upper-right corner of the model. A refined mesh was applied around the defect, with a minimum element size of 0.0625 mm, to enable accurate calculation of the stress intensity field. The resulting simulation reproduced the observed delamination behavior and calculated the total strain energy release rate along the crack front, as shown in Fig. 32. After excluding endpoint singularities, the average total strain energy release rate was determined to be  $7.72 \text{ J/m}^2$ . This value was subsequently used as the interfacial fracture toughness, serving as a critical parameter for evaluating delamination susceptibility in future reliability assessments.

**4.5 Process-Dependent Results of Warpage and Delamination.** During thermal cycling in the manufacturing process, mismatches in CTE among constituent materials lead to structural warpage. To ensure planarity following substrate debonding, a flattening procedure is employed; however, this introduces residual stress within the package. Excessive warpage can result in mechanical deformation, which may compromise internal circuitry and affect electrical performance. Accordingly, minimizing warpage is a key objective in packaging development. Repeated thermal cycles further induce nonuniform stress distributions across different layers, increasing the risk of delamination. While the structure is designed to separate cleanly from the substrate during the release stage, delamination during earlier processing must be avoided. Thus, detailed simulations are conducted to predict



**Fig. 27** Predicted warpage of FOPLP structure II



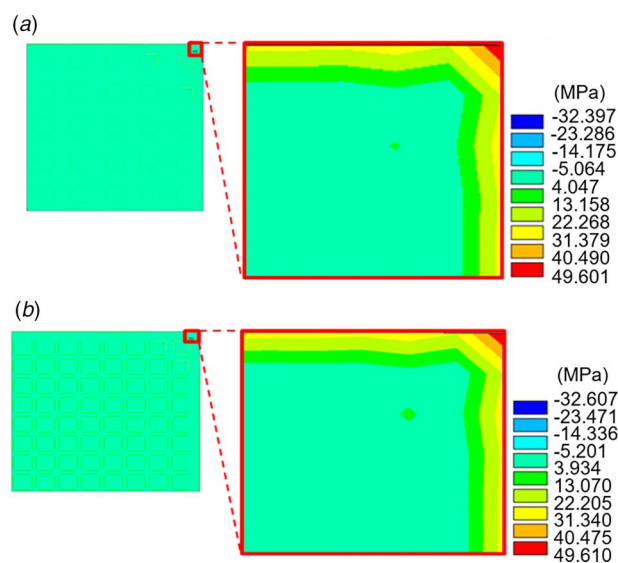
**Fig. 28** Top view of the global FEM model and the zoomed-in corner region of the 1/4 symmetric defect model: (a) die coverage: 10 dies  $\times$  10 dies and (b) die coverage: 16 dies  $\times$  16 dies

mechanical behavior and evaluate process modifications aimed at improving packaging performance.

Die coverage directly influences the number of chips produced per manufacturing cycle. Two die coverage configurations,  $10 \times 10$

**Table 12** Maximum stress at the delamination interface with different die coverage ratios

|               | Die occupation           |                          |
|---------------|--------------------------|--------------------------|
|               | 10 dies $\times$ 10 dies | 16 dies $\times$ 16 dies |
| $\sigma_{zz}$ | 49.60                    | 49.61                    |
| $\sigma_{yz}$ | 17.07                    | 17.24                    |
| $\sigma_{xz}$ | 17.07                    | 17.24                    |



**Fig. 29** z-direction tensile stress distribution at the delamination interface: (a) die coverage: 10 dies  $\times$  10 dies and (b) die coverage: 16 dies  $\times$  16 dies

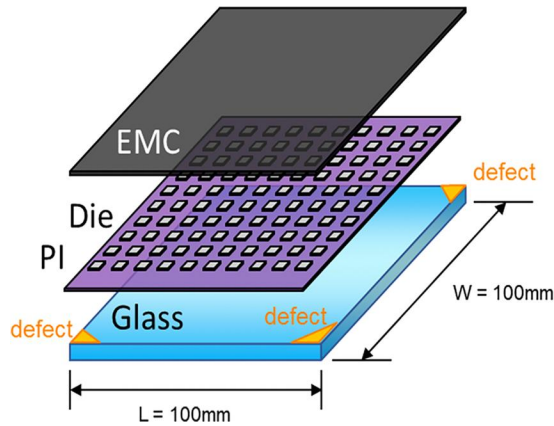


Fig. 30 Schematic of a FOPLP structure with defects

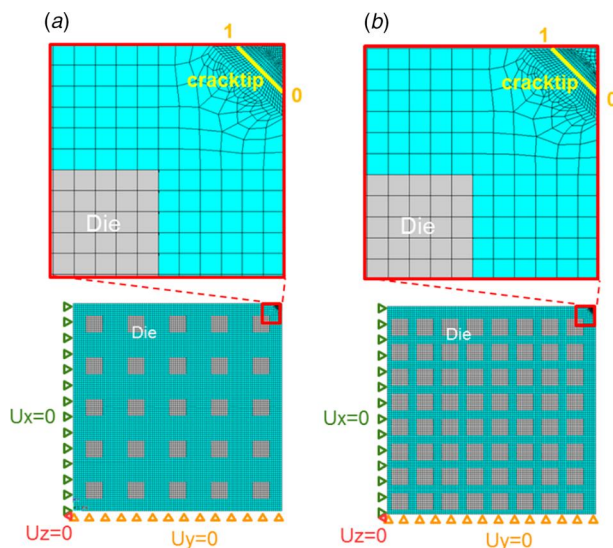


Fig. 31 Top view of the global FEM model and the zoomed-in crack-tip region in the 1/4 symmetric defect model: (a) die coverage: 10 dies  $\times$  10 dies and (b) die coverage: 16 dies  $\times$  16 dies

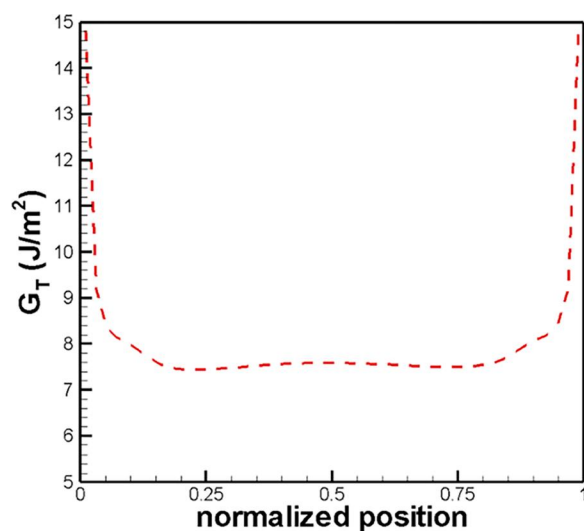


Fig. 32 Fracture toughness of FOPLP structure I

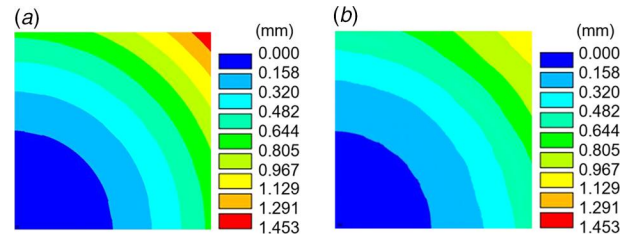


Fig. 33 Predicted warpage for structure I: (a) die coverage 10  $\times$  10 and (b) die coverage 16  $\times$  16

and 16  $\times$  16, were analyzed to evaluate their impact on warpage and delamination. Assuming no delamination after thermal loading, warpage simulations yielded values of 1.45 mm and 1.18 mm for the 10  $\times$  10 and 16  $\times$  16 arrays, respectively, as shown in Fig. 33. The reduced warpage in the 16  $\times$  16 configuration is attributed to the increased number of stiffer die structures, which are more uniformly distributed across the panel.

Delamination analysis was performed using the meshed models presented in Fig. 31. The total strain energy release rate was calculated along the crack front for each configuration, with results shown in Fig. 34. Excluding endpoint singularities, the average total strain energy release rates were 10.08 and 10.10 J/m<sup>2</sup>, both exceeding the interfacial fracture toughness of 7.72 J/m<sup>2</sup>. This indicates that delamination is expected in both scenarios. Moreover, the difference in energy release rate between the two die coverages is negligible, suggesting that while increasing die coverage mitigates warpage, it does not prevent interfacial failure.

The molding process serves to encapsulate the die, providing environmental protection, mechanical shielding, and thermal dissipation. However, the curing of EMC introduces substantial volumetric shrinkage, which is a primary cause of warpage. Accurate assessment of EMC volume is therefore essential. During molding, an EMC sheet is placed over the die array, heated to soften and flow over the structure, and then cured to complete the encapsulation, as illustrated in Fig. 35.

The relevant geometric parameters include the EMC sheet thickness  $t_{ES}$ , die thickness  $t_{Die}$ , and cured EMC thickness  $t_{EMC}$ . As these parameters are interdependent, variations in die or EMC thickness directly influence the simulation geometry. Experimental values for die thickness range from 150  $\mu$ m to 250  $\mu$ m, and EMC sheet thickness ranges from 200  $\mu$ m to 300  $\mu$ m.

In this study, a baseline die thickness of 150  $\mu$ m is adopted for the original configuration (Model I), while variations up to 250  $\mu$ m are

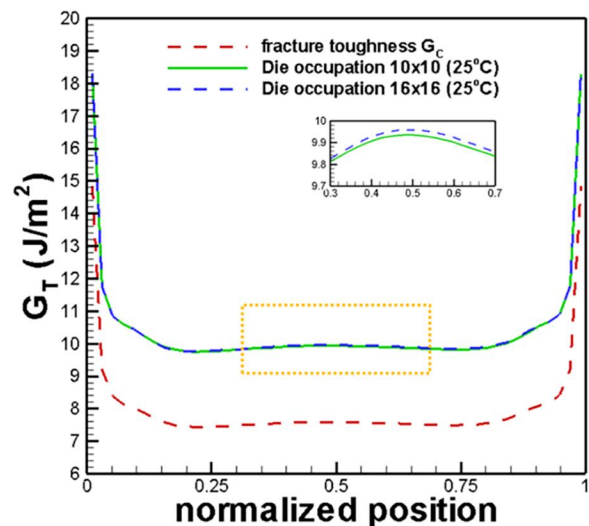
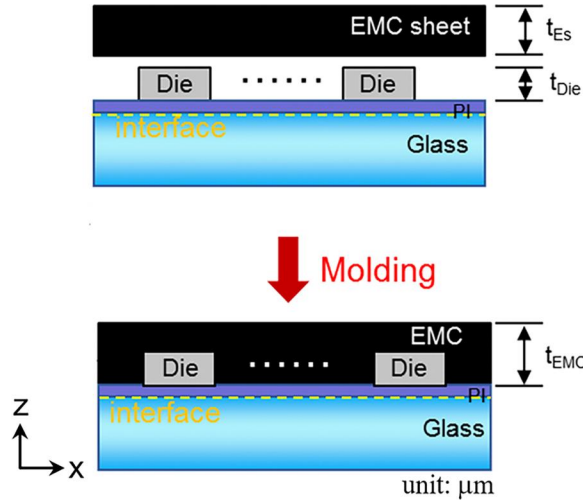


Fig. 34 Total strain energy release rate in defect models with different die coverage levels



**Fig. 35 Schematic of the encapsulation process for the FOPLP structure 1**

also investigated, as shown in Fig. 36. The required EMC sheet thickness is computed based on a simplified volumetric approach, as defined by the following equations:

$$V_{Es} + V_{Die} + V_{error} = V_{EMC} \quad (18)$$

$$t_{Es} \times A_{Es} + t_{Die} \times A_{Die} + V_{error} = t_{EMC} \times A_{EMC} \quad (19)$$

$V$ ,  $A$ , and  $t$  represent volume, die coverage area, and thickness, respectively. To account for process nonidealities not captured in the simplified volumetric calculation (Eq. (19)), such as minor surface unevenness, an empirical volume error ( $V_{error}$ ) was introduced. This term was not directly measured but was calculated using Eq. (19) by determining the residual volume from the experimentally measured geometric parameters. For the baseline  $10 \times 10$  die configuration of FOPLP Structure I, this calculated value was approximately  $0.06 \text{ mm}^3$ . It should be noted that for the analysis of the  $16 \times 16$  configuration, this absolute error value was kept constant as a

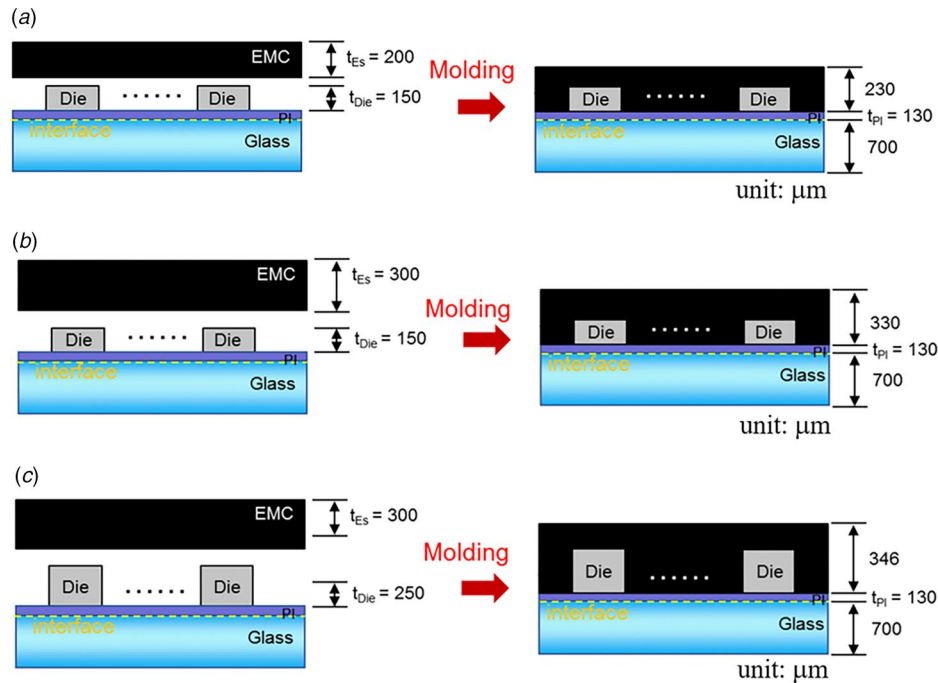
first-order approximation. Schematic of the three models is presented in Fig. 36, where model 1 represents the original configuration of packaging structure I.

Under thermal loading during the manufacturing process, the packaging structure was simulated by progressively cooling to ambient temperature ( $25^\circ\text{C}$ ), assuming ideal interfacial adhesion without delamination. Warpage values for models 1, 2, and 3 were computed as 1.45 mm, 1.82 mm, and 1.81 mm, respectively. The results indicate that increasing the EMC thickness significantly amplifies warpage due to greater volumetric shrinkage during curing. In contrast, increasing the die thickness slightly reduces warpage, attributed to the added stiffness of the thicker die.

To evaluate delamination risk, microdefect simulations were conducted by excluding singularities at the crack-tip ends to ensure accuracy. The average total strain energy release rates for the three models were calculated as 10.08, 10.67, and  $12.38 \text{ J/m}^2$ , respectively, as shown in Fig. 37. The elevated values in models 2 and 3 reflect increased stress concentrations associated with larger EMC volumes. These findings highlight a critical design tradeoff. While a thicker die can reduce warpage due to increased stiffness, it also leads to a higher strain energy release rate, thereby increasing the risk of interfacial failure. Therefore, if minimizing delamination is the primary objective, thinner EMC and die layers are favorable for enhancing the fracture reliability of the package.

Substrate material selection plays a critical role in influencing warpage and delamination behavior during the debonding stage in FOPLP. The substrate directly interfaces with the PI layer through a release coating that facilitates detachment by reducing interfacial adhesion. Therefore, the thermal and mechanical properties of the substrate significantly impact delamination behavior.

To investigate substrate effects, three glass substrates identified as A1\_1 (baseline), A1\_2, and A1\_3 were analyzed under thermal cycling conditions. Their mechanical properties are listed in Table 13, and simulated warpage results are summarized in Table 14. The results show that increasing substrate thickness improves structural rigidity and effectively reduces warpage. Furthermore, aligning the substrate's CTE with the effective laminate CTE helps minimize thermal mismatch and associated deformation. These observations highlight the combined benefit of optimized thickness and matched thermal properties for reducing warpage. Figure 38 illustrates the total strain energy release rates at the delamination interface for



**Fig. 36 Schematic of the EMC and die thicknesses: (a) Model I, (b) Model II, and (c) Model III**

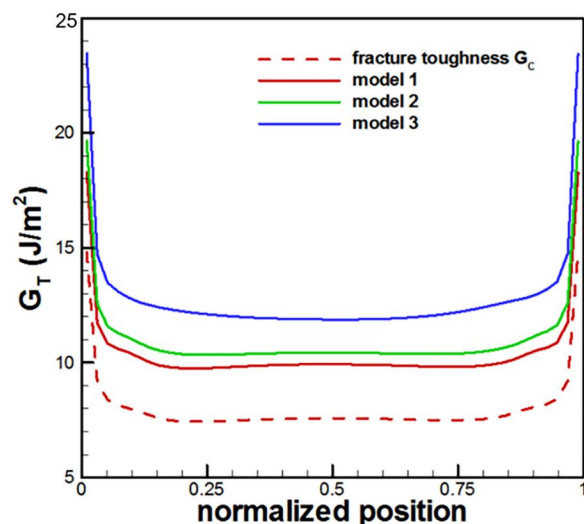


Fig. 37 Total strain energy release rates for defect models with different EMC sheet and die thicknesses

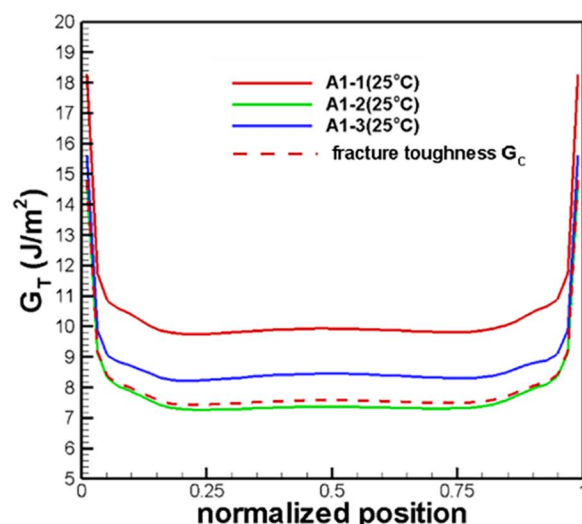


Fig. 38 Total strain energy release rates for defect models with different glass substrate

Table 13 Mechanical properties of various glass substrate

| Glass | Thickness (mm) | $E$ (GPa) | CTE ( $10^{-6}/^{\circ}\text{C}$ ) | $\nu$ |
|-------|----------------|-----------|------------------------------------|-------|
| A1_1  | 0.7            | 73.2      | 3.8                                | 0.3   |
| A1_2  | 0.7            | 73.2      | 5                                  | 0.3   |
| A1_3  | 1.1            | 73.2      | 5                                  | 0.3   |
| D2    | 2.8            | 76        | 8.4                                | 0.3   |
| C1    | 0.7            | 69.3      | 7.9                                | 0.3   |
| C4    | 2.0            | 67.2      | 6.4                                | 0.3   |

varying substrate thicknesses. While increased thickness reduces warpage, it also intensifies gravitational effects, contributing to higher crack-tip stress and elevated energy release rates. Comparisons between A1\_1 and A1\_2 confirm that substrates with CTEs closely matched to the laminate more effectively mitigate shrinkage-induced stresses.

Based on the findings from numerical simulations, four glass substrates were selected for experimental evaluation. These include glass A1, which corresponds to model A1\_1, glass D2 supplied by Dow Chemical Company, and glasses C1 and C4 provided by Corning Incorporated. The mechanical properties of the selected substrates are summarized in Table 13, and the simulated warpage results under thermal cycling conditions are presented in Table 14. The simulation outcomes indicate that substrates with higher coefficients of thermal expansion, such as glass D2 and glass C1, exhibit warpage behavior consistent with theoretical expectations. Furthermore, as shown in Fig. 39, the total strain energy release rate for glass C4 remains below the critical fracture toughness of the release interface, indicating enhanced resistance to delamination. These results demonstrate that both the thermal expansion properties and the mechanical stiffness of the substrate significantly

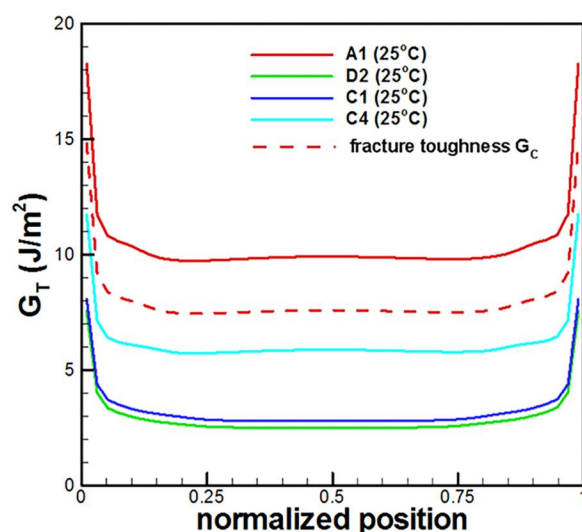


Fig. 39 Total strain energy release rate of defect models with 4 different glass substrates

influence warpage and interfacial reliability. Substrate selection is therefore a critical factor in improving the mechanical performance and durability of FOPLP.

Warpage analysis was conducted on simplified packaging structure I. During thermal loading, the packaging structure exhibited a warpage of 1.45 mm. After laser debonding and removal of the glass substrate, warpage increased to 1.57 mm due to the absence of structural restraint provided by the glass. These results are illustrated in Fig. 40, which presents the deformation profiles

Table 14 Warpage of the FOPLP structure 1 with different glass substrate

| Glass | Warpage (mm) |
|-------|--------------|
| A1_1  | 1.45         |
| A1_2  | 1.21         |
| A1_3  | 1.1          |
| D2    | 0.005        |
| C1    | 0.5          |
| C4    | 0.13         |

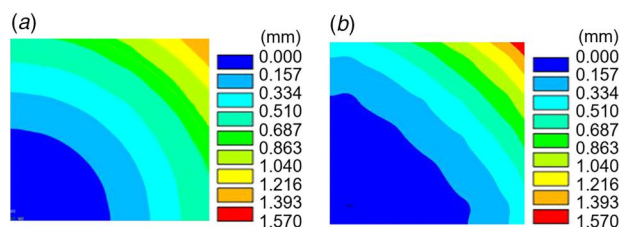


Fig. 40 Predicted warpage for structure I: (a) before debonding and (b) after debonding

before and after substrate removal. This increase in warpage following debonding highlights the critical role of the glass substrate in constraining deformation during the curing and cooling phases. The simulation underscores the need to consider postdebonding mechanical behavior in warpage prediction and reinforces the importance of structural and material optimization to ensure dimensional stability and reliability in panel-level packaging.

## 5 Conclusions

This study developed a multiscale finite element modeling framework to investigate interfacial delamination and warpage behavior in FOPLP. By integrating global and submodels, the proposed approach captures localized stress concentrations near copper pillars and critical interfaces, providing predictive accuracy for crack initiation and delamination during the debonding process. Experimental validation confirmed the accuracy of the simulation results, particularly in warpage prediction, with simulations showing good agreement with measured data. Parametric analyses revealed that increasing UBM chamfer radius, decreasing interfacial fracture toughness, and enlarging UBM etching angles significantly reduce stress concentrations, thereby mitigating delamination risk.

The influence of die coverage, EMC/die thickness, and substrate material on warpage and interfacial fracture behavior was systematically assessed. Higher die coverage and thinner EMC structures were found to reduce warpage, while substrate CTE matching and increased substrate stiffness minimized delamination susceptibility. Among evaluated substrates, glass D2 and glass C1/C4 exhibited favorable performance in suppressing interfacial fracture. The simulation framework also incorporated process-induced effects such as chemical shrinkage and element birth-death techniques to model laser debonding, further enhancing model realism. The findings provide practical guidance for optimizing FOPLP structural design and process parameters to improve reliability and yield in advanced electronic packaging.

## Acknowledgment

The authors also thank the National Center for High-Performance Computing (NCHC) for computational resources support.

## Funding Data

- National Science and Technology Council, Taiwan, ROC (Grant Nos. NSTC 113-2221-E-035-031-MY3 and NSTC 114-2222-E-239-003; Funder ID: 10.13039/501100001868).

## Data Availability Statement

The authors attest that all data for this study are included in the paper.

## References

- [1] Lau, J., Li, M., Tian, D., Fan, N., Kuah, E., Kai, W., Li, M., et al., 2017, "Warpage and Thermal Characterization of Fan-Out Wafer-Level Packaging," *ASME J. Electron. Packag.*, **67**, pp. 595–602.
- [2] Lau, J. H., Li, M., Li, Q. M., Xu, I., Chen, T., Li, Z., Tan, K. H., et al., 2018, "Design, Materials, Process, Fabrication, and Reliability of Fan-Out Wafer-Level Packaging," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **8**(6), pp. 991–1002.
- [3] Katari, M., Krishnamoorthy, G., and Jeyaraman, J., 2024, "Novel Materials and Processes for Miniaturization in Semiconductor Packaging," *J. Artif. Intell. Gen. Sci.*, **2**(1), pp. 251–271.
- [4] Cheng, H. C., Wu, Z. D., and Liu, Y. C., 2020, "Viscoelastic Warpage Modeling of Fan-Out Wafer-Level Packaging During Wafer-Level Mold Cure Process," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **10**(7), pp. 1240–1250.
- [5] Nishimura, I., Yamagami, M., Oji, H., and Hayashi, T., 2020, "RDL-First FOWLP for Low-Density Applications With New Concept FOWLP Technology," Proceedings of International Wafer Level Packaging Conference (IWLP), San Jose, CA, Oct. 13–30, pp. 1–6.
- [6] Yu, C. F., Peng, J. W., Hsiao, C. C., Wang, C. H., and Lo, W. C., 2025, "Development of GUI-Driven AI Deep Learning Platform for Predicting Warpage Behavior of Fan-Out Wafer-Level Packaging," *Micromachines*, **16**(3), p. 342.
- [7] Lee, C. C., Lin, Y. H., and Yang, D. P., 2024, "Stress-Induced Warpage Estimation of Advanced Semiconductor Copper Interconnect Processes," *Int. J. Mech. Sci.*, **284**, p. 109744.
- [8] Lau, J. H., Chen, G. C.-F., Huang, J. Y.-C., Chou, R. T.-S., Yang, C. C.-L., Liu, H.-N., and Tseng, T.-J., 2021, "Hybrid Substrate by Fan-Out RDL-First Panel-Level Packaging," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **11**(8), pp. 1301–1309.
- [9] Lau, J. H., Chen, C. F., Huang, Y. C., Chou, T. S., Yang, C. L., and Liu, H. N., 2021, "Fan-Out (RDL-First) Panel-Level Hybrid Substrate for Heterogeneous Integration," Proceedings of IEEE 71st Electronic Components and Technology Conference (ECTC), San Jose, CA, June 1–4, pp. 1–6.
- [10] Chin, I., Loh, W. K., Kee, S. C., He, Y., Abdullah, M. Z., and Tsuruya, M., 2025, "Experimental Method to Measure High-Temperature Hygroscopic Swelling in Epoxy Mold Compounds and Dielectric Build-Up Films Available to Purchase," *ASME J. Electron. Packag.*, **147**(1), p. 011009.
- [11] Kim, J., Choi, I., Park, J. H., Lee, J. E., Jeong, T. S., Byun, J., Ko, Y. G., Hur, K., Kim, D. W., and Oh, K. S., 2018, "Fan-Out Panel Level Package With Fine Pitch Pattern," Proceedings of 2018 IEEE 68th Electronic Components and Technology Conference (ECTC), San Diego, CA, May 29–June 1, pp. 52–57.
- [12] Ko, C.-T., Yang, H., Lau, J. H., Li, M., Li, M., Lin, C., Lin, J. W., Chen, T., Xu, I., Chang, C.-L., Pan, J.-Y., Wu, H.-H., Yong, Q. X., Fan, N., Kuah, E., Li, Z., Tan, K. H., Cheng, Y.-M., Ng, E., Kai, W., Hao, J., Beica, R., Lin, M., Chen, Y.-H., Cheng, Z., Wee, K. S., Ran, J., Xi, C., Lim, S. P., Lee, N. C., Tao, M., Lo, J., and Lee, R. S. W., 2018, "Chip-First Fan-Out Panel-Level Packaging for Heterogeneous Integration," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **8**(9), pp. 1561–1572.
- [13] Tsai, C. H., Liu, S. W., and Chiang, K. N., 2020, "Warpage Analysis of Fan-Out Panel-Level Packaging Using Equivalent CTE," *IEEE Trans. Device Mater. Reliab.*, **20**(1), pp. 51–57.
- [14] Che, F. X., Yamamoto, K., Rao, V. S., and Sekhar, V. N., 2020, "Panel Warpage of Fan-Out Panel-Level Packaging Using RDL-First Technology," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **10**(2), pp. 304–313.
- [15] Cheng, H. C., and Liu, Y. C., 2020, "Warpage Characterization of Molded Wafer for Fan-Out Wafer-Level Packaging," *ASME J. Electron. Packag.*, **142**(1), p. 011004.
- [16] Dijk, M., Huber, S., Stegmaier, A., Walter, H., Wittler, O., and Schneider-Ramelow, M., 2022, "Experimental and Simulative Study of Warpage Behavior for Fan-Out Wafer-Level Packaging," *Microelectron. Reliab.*, **135**, p. 114585.
- [17] Yu, C. F., Huang, Y. W., Ouyang, T. Y., Cheng, S. F., Chang, H. H., and Hsiao, C. C., 2022, "Suppression Strategy for Process-Induced Warpage of Novel Fan-Out Wafer-Level Packaging," *Microelectron. Reliab.*, **136**, p. 114683.
- [18] Chen, C., Su, M., Ma, R., Zhou, Y., Li, J., and Cao, L., 2022, "Investigation of Warpage for Multi-Die Fan-Out Wafer-Level Packaging Process," *Materials*, **15**(5), p. 1683.
- [19] Lee, C. C., Chang, C. P., and Huang, P. C., 2023, "Development and Demonstration on Process-Oriented Warpage Simulation Methodology of Fan-Out Panel-Level Package in Multilevel Integration," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **13**(12), pp. 2016–2023.
- [20] Braun, T., Höck, O., Adler, M., Obst, M., Voges, S., and Becker, K. F., 2024, "How to Manipulate Warpage in Fan-Out Wafer and Panel-Level Packaging," Proceedings of 2024 IEEE 74th Electronic Components and Technology Conference (ECTC), Denver, CO, May 28–31, pp. 1–6.
- [21] Conversion, A., Ubando, A. T., and Gonzaga, J., 2023, "Interfacial Delamination Validation on Fan-Out Wafer-Level Package Using Finite Element Method," *Solid State Phenom.*, **343**, pp. 73–78.
- [22] Wu, W., Zhao, W., Chen, K., Ma, B., Lu, D., Wang, K., and Wu, J., 2024, "Quantitative Evaluation of PI-RDL Interfacial Delamination in Fan-Out Wafer-Level Packaging During Unbiased Highly Accelerated Stress Test," *Eng. Failure Anal.*, **164**, p. 108668.
- [23] Che, F. X., Ho, D., Ding, M. Z., and MinWoo, D. R., 2016, "Study on Process Induced Wafer Level Warpage of Fan-Out Wafer Level Packaging," Proceedings of 2016 IEEE 66th Electronic Components and Technology Conference (ECTC), Las Vegas, NV, May 31–June 3, pp. 1879–1885.
- [24] Chiu, T. C., and Yeh, E. Y., 2018, "Warpage Simulation for the Reconstituted Wafer Used in Fan-Out Wafer Level Packaging," *Microelectron. Reliab.*, **80**, pp. 14–23.
- [25] Wu, M. L., and Lan, J. S., 2019, "Simulation and Experimental Study of the Warpage of Fan-Out Wafer-Level Packaging: The Effect of the Manufacturing Process and Optimal Design," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **9**(7), pp. 1396–1405.
- [26] Li, W., and Yu, D., 2022, "Experimental and Numerical Investigations of the Warpage of Fan-Out Wafer-Level Packaging With SAW Filters," *IEEE Trans. Compon. Packag. Manuf. Technol.*, **12**(5), pp. 869–877.
- [27] Chuang, W. C., Huang, Y., and Chen, P. E., 2023, "Exploring the Influence of Material Properties of Epoxy Molding Compound on Wafer Warpage in Fan-Out Wafer-Level Packaging," *Materials*, **16**(9), p. 3482.
- [28] Oi, Y., Fujii, Y., Hiraoka, T., Yada, Y., and Kan, K., 2018, "Warpage Control of Liquid Molding Compound for Fan-Out Wafer Level Packaging," Proceedings of 2018 IEEE 68th Electronic Components and Technology Conference (ECTC), San Diego, CA, May 29–June 1, pp. 967–972.
- [29] Li, H. Y., Chen, A., Peng, S., Pan, G., and Chen, S., 2017, "Warpage Tuning Study for Multi-Chip Last Fan-Out Wafer Level Package," Proceedings of 2017 IEEE 67th Electronic Components and Technology Conference (ECTC), Orlando, FL, May 30–June 2, pp. 1384–1391.
- [30] Zhu, C., Guo, P., and Dai, Z., 2017, "Investigation on Wafer Warpage Evolution and Wafer Asymmetric Deformation in Fan-Out Wafer Level Packaging

- Processes,” Proceedings of 2017 18th International Conference on Electronic Packaging Technology (ICEPT), Harbin, China, Aug. 16–19, pp. 664–668.
- [31] Irwin, G. R., 1957, “Analysis of Stresses and Strains Near the End of a Crack Traversing a Plate,” *ASME J. Appl. Mech.*, **24**(3), pp. 361–364.
- [32] Irwin, G. R., 1968, “Linear Fracture Mechanics, Fracture Transition, and Fracture Control,” *Eng. Fract. Mech.*, **1**(2), pp. 241–257.
- [33] Irwin, G. R., and Wells, A. A., 1965, “A Continuum-Mechanics View of Crack Propagation,” *Int. Mater. Rev.*, **10**(1), pp. 223–270.
- [34] Rybicki, E. F., and Kanninen, M. F., 1977, “A Finite Element Calculation of Stress Intensity Factors by a Modified Crack Closure Integral,” *Eng. Fract. Mech.*, **9**(4), pp. 931–938.
- [35] Shivakumar, K. N., Tan, P. W., and Newman, J. C., 1988, “A Virtual Crack-Closure Technique for Calculating Stress Intensity Factors for Cracked Three-Dimensional Bodies,” *Int. J. Fract.*, **36**(3), pp. R43–R50.
- [36] Okada, H., Higashi, M., Kikuchi, M., Fukui, Y., and Kumazawa, N., 2005, “Three Dimensional Virtual Crack Closure-Integral Method (VCCM) With Skewed and Non-Symmetric Mesh Arrangement at the Crack Front,” *Eng. Fract. Mech.*, **72**(11), pp. 1717–1737.
- [37] Smith, S. A., and Raju, I. S., 1999, “Evaluation of Stress-Intensity Factors Using General Finite-Element Models,” *Fatigue and Fracture Mechanics: 29th Volume*, ASTM International, West Conshohocken, PA.
- [38] Cho, J., Joshi, M., and Sun, C. T., 2006, “Effect of Inclusion Size on Mechanical Properties of Polymeric Composites With Micro and Nano Particles,” *Compos. Sci. Technol.*, **66**(13), pp. 1941–1952.
- [39] Sun, C. T., and Jih, C., 1987, “On Strain Energy Release Rates for Interfacial Cracks in Bi-Material Media,” *Eng. Fract. Mech.*, **28**(1), pp. 13–20.
- [40] Sun, C. T., and Qian, W., 1997, “The Use of Finite Extension Strain Energy Release Rates in Fracture of Interfacial Cracks,” *Int. J. Solids Struct.*, **34**(20), pp. 2595–2609.
- [41] Raju, I. S., 1987, “Calculation of Strain-Energy Release Rates With Higher Order and Singular Finite Elements,” *Eng. Fract. Mech.*, **28**(3), pp. 251–274.
- [42] Murthy, D. P. V., and Raju, I. S., 1998, “Modified Virtual Crack Closure Integral (MVCCI) for Calculating Stress Intensity Factors and Energy Release Rates in Anisotropic Materials,” *Compos. Struct.*, **41**(3–4), pp. 333–341.
- [43] Alfano, G., and Crisfield, M. A., 2001, “Finite Element Interface Models for the Delamination Analysis of Laminated Composites: Mechanical and Computational Issues,” *Int. J. Numer. Methods Eng.*, **50**(7), pp. 1701–1736.
- [44] Krueger, R., 2004, “The Virtual Crack Closure Technique (VCCT) in Engineering Practice,” *ASME Appl. Mech. Rev.*, **57**(2), pp. 109–143.
- [45] Xie, D., and Biggers, S. B., 2006, “Progressive Crack Growth Analysis Using a New Multi-Point Virtual Crack Closure Technique,” *Finite Elem. Anal. Des.*, **42**(11), pp. 977–984.
- [46] Zaghi, A. E., and Shah, A., 2014, “A Review of the Virtual Crack Closure Technique for the Calculation of the Strain Energy Release Rate,” *J. Stress Anal.*, **1**(1), pp. 1–10.
- [47] Gibson, R. F., 2016, *Principles of Composite Material Mechanics*, CRC Press, Boca Raton, FL.
- [48] Sørensen, B. F., Sarraute, S., Jørgensen, O., and Horswell, A., 1998, “Thermally Induced Delamination of Multilayers,” *Acta Mater.*, **46**(8), pp. 2603–2615.
- [49] Budzik, M. K., Heide-Jørgensen, S., and Aghababaei, R., 2021, “Fracture Mechanics Analysis of Delamination Along Width-Varying Interfaces,” *Compos. Part B Eng.*, **215**, p. 108793.